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MIL-M-28787/536
15 SEPTEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY INTERFACE II MODULE, KEY CODE EWD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory interface II module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.287 failures per million operating hours for class I or 3.029 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be EWD.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A4	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D2-D6	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D7	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D8-D11	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D12,D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D14	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D15-D20	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D21,D22	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D23	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D24	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D25	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D26-D33	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D34	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D35-D38	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D39,D40	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D41-D48	Ibd	V	S	22.93	27.32	22.93	27.32	22.93	27.32	mA
D49	Ibd	V	S	22.93	29.32	22.93	29.32	22.93	29.32	mA
D50-D57	Ibd	V	S	22.93	27.32	22.93	27.32	22.93	27.32	mA
E1-E8	Ios	VI	S	-225.0	-100.0	-225.0	-100.0	-225.0	-100.0	mA
E9-E12	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E13-E22	Ios	VI	S	-225.0	-100.0	-225.0	-100.0	-225.0	-100.0	mA
E23	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E24-E27	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E28-E30	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F2-F5	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F6	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F7	Ith	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F8-F11	Ith	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F12,F13	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F14	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F15-F20	Ith	VII	X	-0.8	0.02	-0.8	0.02	-0.8	0.02	mA
F21	Ith	VII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
F22	Ith	VII	X	-0.6	0.06	-0.6	0.06	-0.6	0.06	mA
F23	Ith	VII	X	-7.23	-2.85	-7.23	-2.85	-7.23	-2.85	mA
F24	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F25	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F26-F33	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F34	Ith	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F35-F38	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F39,F40	Ith	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F41-F48	Ith	VII	X	-28.26	-4.36	-28.26	-4.36	-28.26	-4.36	mA
F49	Ith	VII	X	-35.06	-4.21	-35.06	-4.21	-35.06	-4.21	mA
F50-F57	Ith	VII	X	-28.26	-4.36	-28.26	-4.36	-28.26	-4.36	mA
G1	Itl	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G2-G5	Itl	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G6	Itl	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G7	Itl	VIII	X	-3.8	0.11	-3.8	0.11	-3.8	0.11	mA
G8-G11	Itl	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G12,G13	Itl	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G14	Itl	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G15-G20	Itl	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G21	Itl	VIII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
G22	Itl	VIII	X	-2.6	0.11	-2.6	0.11	-2.6	0.11	mA
G23	Itl	VIII	X	-7.23	-2.85	-7.23	-2.85	-7.23	-2.85	mA
G24	Itl	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G25	Itl	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G26-G33	Itl	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G34	Itl	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
G35-G38	Itl	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G39,G40	Itl	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G41-G48	Iil	VIII	X	-28.26	-4.36	-28.26	-4.36	-28.26	-4.36	mA
G49	Iil	VIII	X	-35.06	-4.21	-35.06	-4.21	-35.06	-4.21	mA
G50-G57	Iil	VIII	X	-28.26	-4.38	-28.26	-4.38	-28.26	-4.36	mA
H1-H30	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J30	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K18	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K19-K36	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L2-L5	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L6,L7	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L8-L11	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L12-L21	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L22,L23	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L24	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L25-L36	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XIII	S	3.6	6.0	3.6	6.0	3.6	6.0	uF
M2	PSFC	XIII	S	4.6	8.7	4.6	8.7	4.6	8.7	uF
N	Icc	XIV	X	590	1416	590	1416	590	1416	mA
P1,P2	Functional	XV	X	See Table XV						
Q1,Q2	Dynamic	XVI	S	See Table XVI						
Q3,Q4	Dynamic	XVI	Q	See Table XVI						
R1-R8	Cin	XVIII	Q	2	50	2	50	2	50	pF
R9	Cin	XVIII	Q	2	75	2	75	2	75	pF

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R10,R11	Cin	XVIII	Q	2	50	2	50	2	50	pF
R12,R13	Cin	XVIII	Q	4	75	4	75	4	75	pF
R14	Cin	XVIII	Q	4	50	4	50	4	50	pF
R15-R19	Cin	XVIII	Q	4	75	4	75	4	75	pF
R20-R28	Cin	XVIII	Q	2	30	2	30	2	30	pF
R29	Cin	XVIII	Q	4	100	4	100	4	100	pF
R30	Cin	XVIII	Q	8	125	8	125	8	125	pF
R31-R34	Cin	XVIII	Q	2	50	2	50	2	50	pF
R35	Cin	XVIII	Q	2	75	2	75	2	75	pF
R36	Cin	XVIII	Q	6	75	6	75	6	75	pF
R37	Cin	XVIII	Q	8	150	8	150	8	150	pF
R38	Cin	XVIII	Q	2	30	2	30	2	30	pF
R39	Cin	XVIII	Q	2	75	2	75	2	75	pF
R40	Cin	XVIII	Q	4	100	4	100	4	100	pF
S	Input clamp diode latch up	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1	PSTR	XX	Q	0	450	0	450	0	450	mV
T2	PSTR	XX	Q	0	650	0	650	0	650	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/536-1 for class I and M28787/536-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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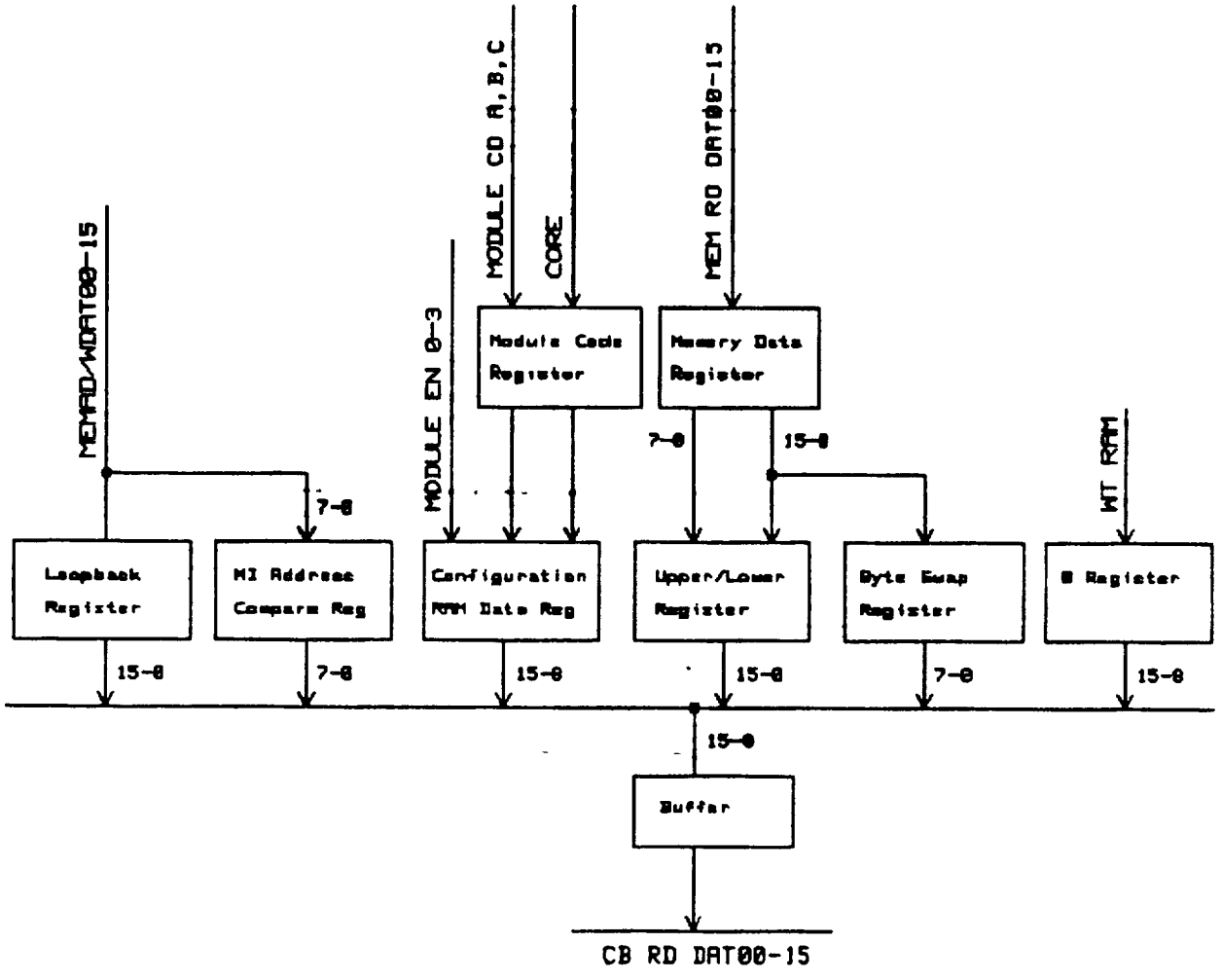


FIGURE 1. EWD functional block diagram.

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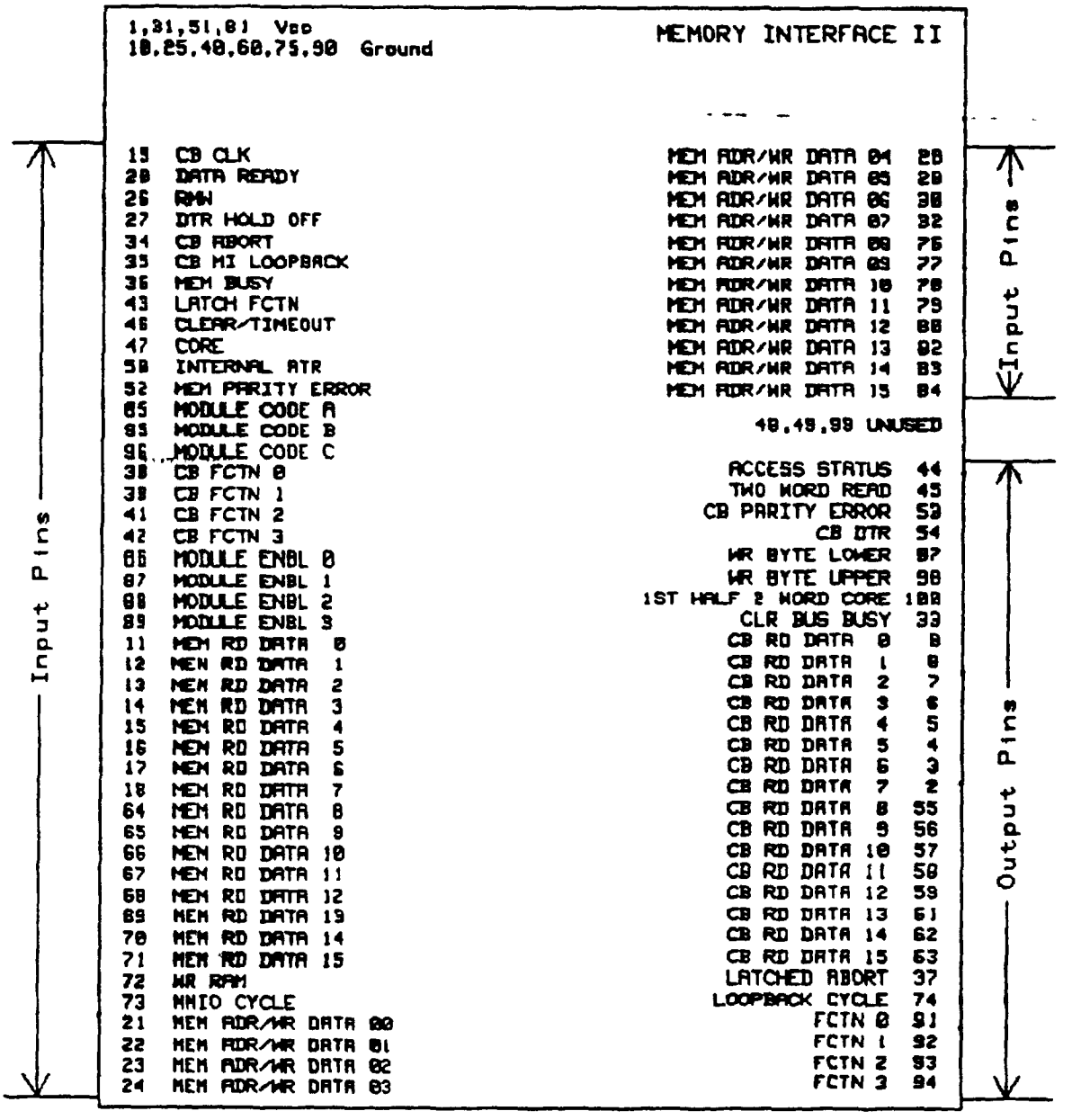


FIGURE 2. EWD pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L and S where applicable, followed by the remaining tests in the sequence.

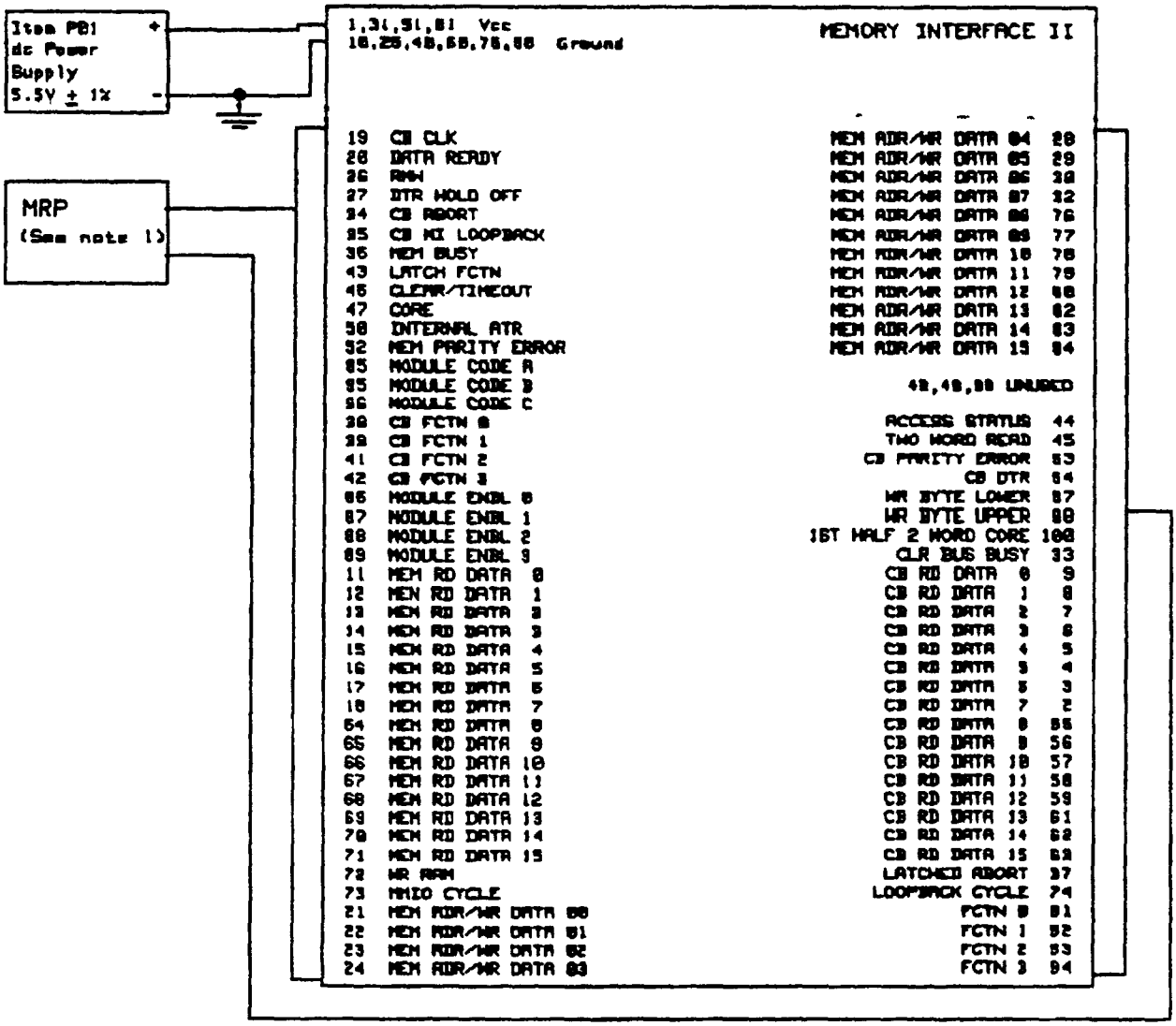
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. EWD operating temperature and component temperature test circuit.

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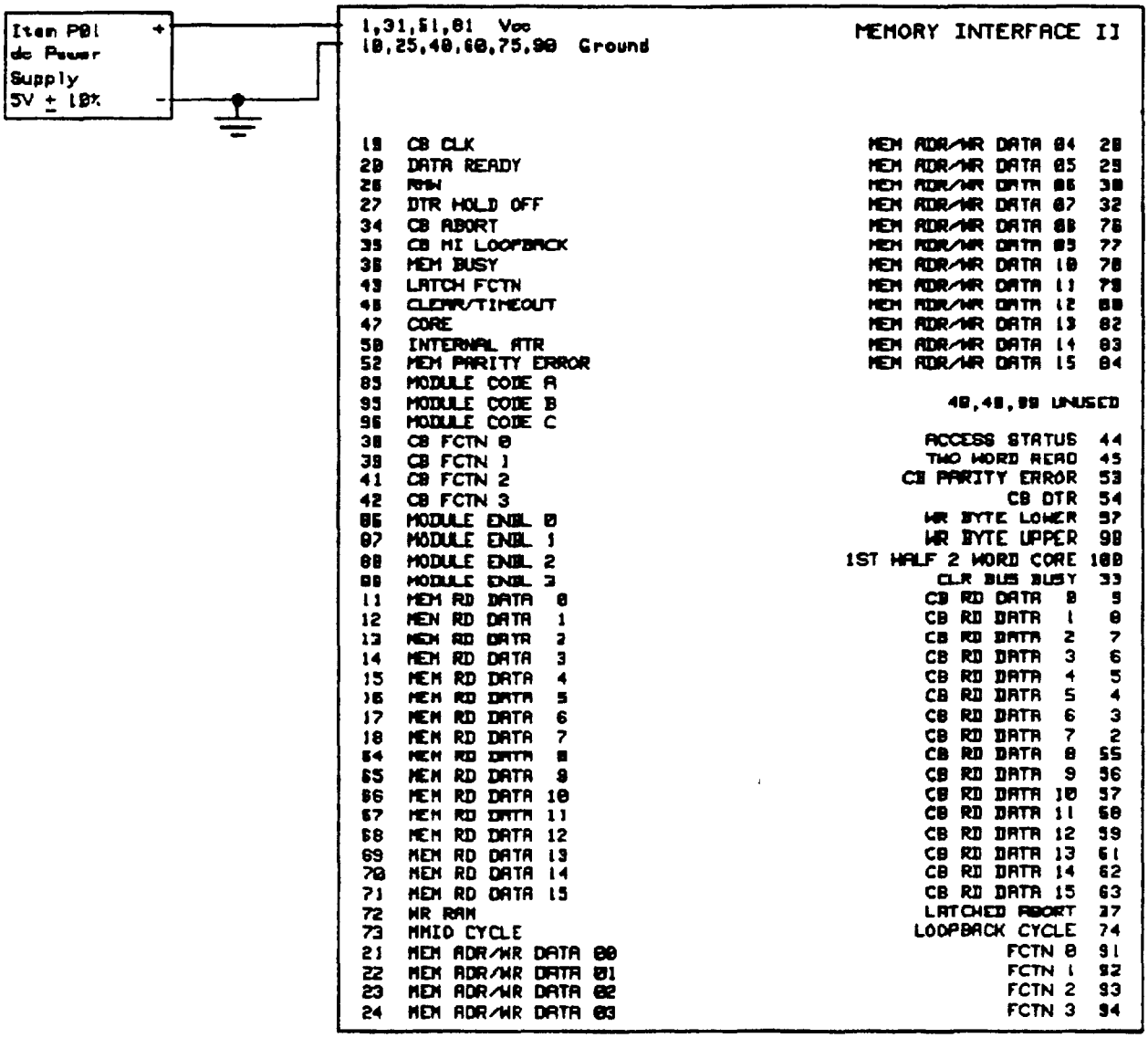


FIGURE 4. EWD vibration test circuit.

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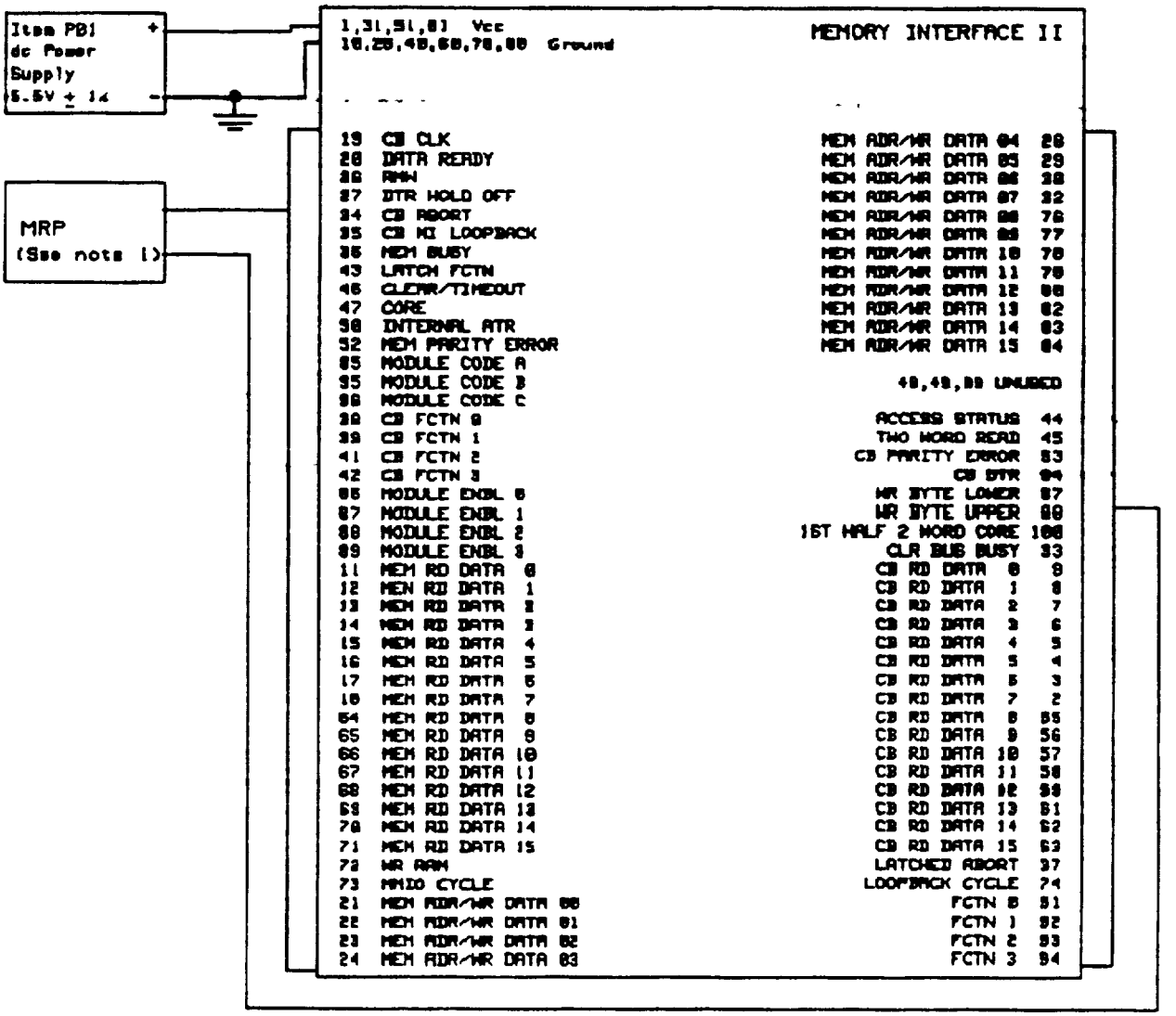
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. EWD life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) Iozl = output low level leakage current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 7.79 W (max); 5.22 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	50
MEMAD/WDAT00	-1.5	0.4	2.4	-0.4	0.04	0.2	75
MEMAD/WDAT01	-1.5	0.4	2.4	-0.4	0.04	0.2	75
MEMAD/WDAT02	-1.5	0.4	2.4	-0.4	0.04	0.2	50
MEMAD/WDAT03	-1.5	0.4	2.4	-0.4	0.04	0.2	75
RMW	-1.2	0.4	2.4	-1.2	0.04	0.2	100
DTR HLD OFF	-1.2	0.4	2.4	-3.8	0.06	0.3	125
MEMAD/WDAT04	-1.5	0.4	2.4	-0.4	0.04	0.2	75
MEMAD/WDAT05	-1.5	0.4	2.4	-0.4	0.04	0.2	75
MEMAD/WDAT06	-1.5	0.4	2.4	-0.4	0.04	0.2	75
MEMAD/WDAT07	-1.5	0.4	2.4	-0.4	0.04	0.2	75
CB ABRT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB MI LPBK	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM BSY	-1.2	0.4	2.4	-2.0	0.05	1.0	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB FCTN 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB FCTN 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB FCTN 2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB FCTN 3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
LTCH FCTN	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CLR/TMOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CORE	-1.2	0.4	2.4	-1.4	0.06	0.3	75
INTRNL ATR	-1.5	0.4	2.4	-2.6	0.06	0.3	150
MEM PAR ERR	NA	0.4	2.4	-7.23	-2.85	0.22	50
WT RAM	-1.5	0.4	2.4	-0.2	0.02	0.1	75
MMIO CYC	-1.2	0.4	2.4	-1.2	0.04	0.2	100
MEMAD/WDAT08	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MEMAD/WDAT09	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MEMAD/WDAT10	-1.5	0.4	2.4	-0.2	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEMAD/DAT11	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MEMAD/DAT12	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MEMAD/DAT13	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MEMAD/DAT14	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MEMAD/DAT15	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MODULE CD A	NA	0.4	2.4	-5.63	-2.85	0.22	75
MODULE EN 0	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MODULE EN 1	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MODULE EN 2	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MODULE EN 3	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MODULE CD B	NA	0.4	2.4	-5.63	-2.85	0.22	50
MODULE CD C	NA	0.4	2.4	-5.63	-2.85	0.22	50
MEM RD DAT00	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT01	NA	0.4	2.4	-28.26	-4.36	27.32	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM RD DAT02	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT03	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT04	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT05	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT06	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT07	NA	0.4	2.4	-28.26	-4.36	27.32	NA
DAT RDY	NA	0.4	2.4	-35.06	-4.21	29.32	NA
MEM RD DAT08	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT09	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT10	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT11	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT12	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT13	NA	0.4	2.4	-28.26	-4.36	27.32	NA
MEM RD DAT14	NA	0.4	2.4	-28.26	-4.36	27.32	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM RD DAT15	NA	0.4	2.4	-28.26	-4.36	27.32	NA

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)
CB RD DAT07	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT06	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT05	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT04	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT03	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT02	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT01	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT00	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CLR BUS BUSY	2.4	0.5	-0.4	6.0	NA	NA	-60.0
LTCH ABRT	2.4	0.5	-0.4	6.0	NA	NA	-60.0
ACC STS	2.4	0.5	-0.4	6.0	NA	NA	-60.0
2 WRD RD	2.4	0.5	-0.4	6.0	NA	NA	-60.0
CB PAR ERR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB DTR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)
CB RD DAT08	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT09	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT10	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT11	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT12	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT13	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT14	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
CB RD DAT15	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0
LPBX CYC	2.4	0.5	-0.4	6.0	NA	NA	-60.0
FCTN 0	2.4	0.5	-0.4	6.0	NA	NA	-40.0
FCTN 1	2.4	0.5	-0.4	6.0	NA	NA	-40.0
FCTN 2	2.4	0.5	-0.4	6.0	NA	NA	-40.0
FCTN 3	2.4	0.5	-0.4	6.0	NA	NA	-40.0
WT BYTE LOWR	2.4	0.5	-0.4	6.0	NA	NA	-60.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)
WT BYTE UPPR	2.4	0.5	-0.4	6.0	NA	NA	-60.0
1ST HF 2WD	2.4	0.5	-0.4	6.0	NA	NA	-60.0

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	LTCH FCTN to FCTN 0	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to FCTN 1	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to FCTN 2	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to FCTN 3	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to LPBX FCTN	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to LTCH ABRT	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to WT BYTE LOWR	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to WT BYTE UPPR	-	-	-	30	30	30	nS	12
2	tpd	CB CLK to CB PAR ERR	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT07	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT06	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT05	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT04	-	-	-	40	40	40	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	CB CLK to CB RD DAT03	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT02	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT01	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT00	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT08	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT09	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT10	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT11	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT12	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT13	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT14	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT15	-	-	-	40	40	40	nS	12
3	tsu	MEMAD/WDAT00 to WT RAM	30	30	30	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tsu	MEMAD/WDAT01 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT02 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT03 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT04 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT05 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT06 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT07 to WT RAM	30	30	30	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT00	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT01	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT02	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT03	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT04	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT05	4	4	4	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	th	WT RAM to MEMAD/WDAT06	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT07	4	4	4	-	-	-	nS	12
5	tsu	MODULE EN 0 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MODULE EN 1 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MODULE EN 2 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MODULE EN 3 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT08 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT09 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT10 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT11 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT12 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT13 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT14 to INTRNL ATR	15	15	15	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	MEMAD/WDAT15 to INTRNL ATR	15	15	15	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 0	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 1	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 2	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 3	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT08	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT09	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT10	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT11	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT12	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT13	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT14	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT15	9	9	9	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEM RD DAT00 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT01 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT02 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT03 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT04 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT05 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT06 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT07 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT08 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT09 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT10 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT11 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT12 to DAT RDY	15	15	15	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEM RD DAT13 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT14 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT15 to DAT RDY	15	15	15	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT00	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT01	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT02	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT03	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT04	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT05	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT06	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT07	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT08	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT09	9	9	9	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	DAT RDY to MEM RD DAT10	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT11	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT12	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT13	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT14	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT15	9	9	9	-	-	-	nS	12
9	tsu	MODULE CD A to MEM BSY	15	15	15	-	-	-	nS	12
9	tsu	MODULE CD B to MEM BSY	15	15	15	-	-	-	nS	12
9	tsu	MODULE CD C to MEM BSY	15	15	15	-	-	-	nS	12
10	th	MEM BSY to MODULE EN A	9	9	9	-	-	-	nS	12
10	th	MEM BSY to MODULE EN B	9	9	9	-	-	-	nS	12
10	th	MEM BSY to MODULE EN C	9	9	9	-	-	-	nS	12

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6.4 Module function. The memory interface provides the interface between the common bus and the memory modules. The memory interface is contained on two modules; Memory Interface I (CWG) and Memory Interface II (EWD). Each memory interface provides the interface for a maximum of one million words of magnetic core or semiconductor memory to the AN/UYK-44 common bus. Modularity of memory may be 64K, 128K, 256K, or 512K words with a maximum of four memory modules per memory interface. The maximum UYK-44 memory configuration is four memory interfaces. The memory interface responds to common bus function codes from the common bus master Central Processor (CP) or an Input/Output Controller (IOC) to perform memory transfers or internal transfers. The memory interface consists of an address comparator, address register, write data registers, read data registers, configuration Random Access Memory (RAM), loopback register, Memory Interface (MI) compare register, configuration RAM data register, and control circuits. The address comparator determines if the memory interface is being addressed by comparing common bus address bits 21-15 with the strap bits. Strap bits define the address boundary for each memory interface. When an address is within the defined range, the common bus address and function code are captured. The address register holds the 22-bit address. For memory operations, bits 17-0 are supplied to memory and the upper bits are used to address the configuration RAM which enables the proper module. The function code is translated and the designated operation is performed. Since each memory interface may be configured with a variety of size and type of memory modules, each memory interface is initialized during the CP power-up master clear microprogram routine. The initialization subroutine determines the memory module size and type connected to each memory interface, and then sets up each memory interface configuration RAM to enable each memory module for the proper address range. The configuration RAM contains 16 4-bit words. The words are addressed by the upper address bits. Each bit is associated with a memory module, for example, bit 0 enables module 0. Each RAM word is used to address 32K of memory by enabling the proper module. For example, if a memory module contained 128K words, four consecutive RAM words with the corresponding bit set would be required to address the module. The 16 RAM words address a maximum of 512K. The MI address compare register and configuration RAM data register are used during the initialization subroutine to hold data that determines the memory interface address range and memory module size and type. The write data registers are used to hold data from the common bus. The read data registers hold data from memory. The loopback register provides a means to test the memory interface without using memory by holding the address or write data. Built-In-Test (BIT) control is used by the maintenance processor to detect the memory interface.

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6.4.1 Memory interface initialization. Each memory interface is initialized during the CP power-up master clear routine. The subroutine performs the following sequence of events.

(a) Disable all memory modules and set each MI address compare register to the maximum address associated with each memory interface. The MI address compare register holds the cumulative number of 32K segments to define the upper address boundary for each memory interface. The address is incremented by 32K on each pass. 128 passes are made to address 8 memory interfaces of 512K each (4096K total).

(b) Starting at address zero, determine if a memory interface is present. If not present, increment address by 32K and repeat.

(c) If present, enable module 0 and read the module code from memory to determine module type and size. Write configuration RAM to reflect module size (bit 0 set at each word location for each 32K). The address is incremented by 32K for each configuration RAM write.

(d) Interrogate memory modules 1, 2, and 3 in the same manner as module 0 and set the corresponding module enables in the configuration RAM. If a module is not present, a disable is written into the configuration RAM for that module.

(e) Events (b), (c) and (d) are repeated for each memory interface (except the address is cumulative).

6.4.2 Memory transfers. The memory interface performs the following types of memory data transfers.

(a) Write full word.

(b) Write byte lower.

(c) Write byte upper.

(d) Read full word. (The memory interface modifies the word if the function code specifies a read byte lower or upper by placing the byte in the lower byte position and inserting zeros in the upper byte.)

(e) Read/modify/write.

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The memory interface performs these transfers with four signals to the addressed memory module: Cycle Initiate, Write Byte Lower, Write Byte Upper, and Read/Modify/Write (RMW).

6.4.2.1 Read operations. The memory interface performs a read operation as follows:

(a) Places address on memory address/write data bus and sets Cycle Initiate line high.

(b) The memory responds with the Busy signal indicating it has received the address.

(c) The memory interface clears the Cycle Initiate.

(d) The memory places data on the read data bus and sends the Data Ready signal.

(e) The memory interface accepts the data and places it on the common bus.

For a Nondestructive Readout (NDRO) read operation, the NDRO Enable signal is sent instead of the Cycle Initiate.

6.4.2.2 Write operations. The memory interface performs a write operation as follows:

(a) Places address on memory address/write data bus and sets Cycle Initiate line high, RMW line low, and Write Byte Lower and Write Byte Upper lines depending on operations (when respective line is low, byte is written).

(b) The memory responds with the Busy signal.

(c) The memory interface clears the Cycle Initiate, places the write data on the memory address/write data bus and sets the RMW line high.

(d) The memory senses the high going RMW line and accepts the data.

6.4.2.3 Read/modify/write operations. The memory interface performs a read/modify/write operation as follows:

(a) Places address on memory/write data bus and sets Cycle Initiate line high, RMW, Write Byte Lower, and Write Byte Upper lines low.

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(b) The memory responds with the Busy signal.

(c) The memory interface clears the Cycle Initiate.

(d) The memory places data on the read data bus and sends the Data Ready signal.

(e) The memory interface accepts the data and places it on the common bus.

(f) The memory waits until it receives the modified data from the bus master and then places the write data on the memory address/write data bus and sets the RMW line high.

(g) The memory senses the high going RMW line and accepts the data.

6.4.3 Address comparator. The address comparator compares common bus address bits 21-15 with MI strap bits 21-15 to determine if the address is within the allocated address range for this memory interface. The strap bits are wired externally to +5 volts, ground, and address bits to define the address range for the memory interface. When a match occurs the address is latched in the memory address register.

6.4.4 Memory address register. The 22-bit register holds the memory address during read and write operations. The address from the common bus is latched when the address comparator detects an address range match. Prior to being latched, memory address register bits 21-16 appear transparent to the address comparator. Memory address register bits 15-0 are placed on the memory address/write data bus to memory. Bit 0 passes through a toggle circuit and is unaltered for all operations except a double word reference to core memory. For this operation, the bit is complemented on the first memory reference and is not altered on the second memory reference. Memory address register bits 18-15 are used to address the configuration RAM.

6.4.5 Write data register. The 16-bit register holds memory write data. The register is loaded from the common bus when the data is placed on the bus by the bus master (CP or IOC), the Originator Ready (OR) signal is received, and the function code specifies a write operation (F2 = low). The register output is placed on the memory address/write data bus when the memory Busy signal is received.

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6.4.6 Write byte swap register. The 8-bit register transfers the lower byte (bits 7-0) from the common bus to the upper byte (bits 15-8) sent to memory. The register is loaded with common bus bits 7-0 when the OR signal is received and the function code specifies a write operation (F2 = low). The register output is placed in memory address/write data bus bits 15-8 when the memory Busy signal is received and the function code specifies write byte upper (F2 = low, F1 = high, and F0 = low).

6.4.7 Configuration RAM. The configuration RAM allows the memory interface to interface with a variety of memory module sizes and still maintain addressing continuity. The RAM contains 16 4-bit words. Each bit provides a Module Enable signal to the corresponding memory module (bit 0 enables module 0, bit 1 enables module 1, and so forth). Only one module enable can be active, therefore, only one bit in each word can be set. The RAM is addressed by the upper address bits. Addressing is variable, however, when address bits 18 through 15 are used, each RAM location is used to address 32K of memory. Therefore, the 16 RAM words can address a maximum of 512K (K = 1024 words) of memory. Memory modules can contain 32K, 64K, or 128K words. The configuration RAM is loaded as part of a power-up routine to reflect the module address range. For example, if module 0 contained 64K words and module 1 contained 32K words, bit 0 in RAM words 0 and 1 would be set to enable module 0 for the 0-64K address range and bit 1 in RAM word 1 would be set to enable module 1 for the 64K to 96K address range. The addressed RAM location is loaded with write data register bits 11-8 by a write configuration RAM function code.

6.4.8 Core detector. The core detector provides a Core signal when a memory reference is made to a module containing core memory. The detector consists of a 4-bit register (each bit indicates if the corresponding module contains core) and a compare circuit that compares module enables 3-0 with the corresponding register bits 3-0. The register is loaded with write data register bits 15-21 by a write configuration RAM function code (loaded same time as configuration RAM).

6.4.9 Memory data register. The 16-bit register holds the data read from memory. The register is loaded when the Data Ready signal is received from memory or an internal resume is generated.

6.4.10 Upper/lower register. The 16-bit register holds the memory data word and is loaded from the memory data register when the Data Ready signal from memory is dropped. The register

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is divided into an upper byte (bits 15-8) and a lower byte (bits 7-0). For a read full word function code, both bytes are placed on the common bus via the buffer. For a read byte lower function code, the lower byte is placed on the common bus (the upper byte is zero filled by the 0 register).

6.4.11 Byte swap register. The 8-bit register holds the memory data upper byte and is loaded from the memory data register when the Data Ready signal from memory is dropped. The register output is enabled by a read byte upper function code and is placed on the common bus in the lower byte position (the upper byte is zero filled by the 0 register).

6.4.12 0 register. The 8-bit register zero fills the read data upper byte. The register output is enabled by a read byte lower or a read byte upper function code.

6.4.13 Loopback register. The 16-bit loopback register provides a means to test the memory interface. The register is loaded via the memory address/write data bus from either the memory register or write data register. The address is loaded from read function codes and write data is loaded for write function codes. The register output is enabled and placed on the common bus by the MI Loopback signal from the maintenance processor.

6.4.14 Module code register. The 4-bit register holds the 3-bit module code from the enabled memory module and the Core signal from the core detector. The register is loaded by the memory Busy signal from memory.

6.4.15 Configuration RAM data register. The 8-bit register holds configuration RAM data. The data is read from the register by a read configuration RAM function code and is used as the upper byte (bits 15-8) of the word format placed on the common bus. The register is loaded from the module code register and configuration RAM when the memory interface is addressed.

6.4.16 MI address compare register. The 8-bit register holds configuration RAM data. The register is used during the memory initialization routine to record the number of 32K address segments which are used to determine the memory interface address range. The register is loaded from the memory address/write data bus bits 7-0 by a write configuration RAM function code. The data is read from the register by a read configuration RAM function code and is used as the lower byte (bits 7-0) of the word format placed on the common bus.

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6.4.17 Function decoder. The function decoder translates the 4-bit common bus function code to provide the memory interface control signals. The function code is latched when the address comparator detects an address range match.

6.4.18 BIT control. BIT control provides the maintenance processor (BWH) with information. The maintenance processor activates BIT control with the CB MI Enable signal. BIT control responds with the MI Present signal (low). It also responds with the MI Address Compare signal (low) if the memory interface has been addressed via the common bus.

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6.4.19 Description of I/O pins.

Signal name	Pin number	I/O	Description
CB CLK	19	I	CB CLK
MEMAD/WDAT00	21	I	MEM ADR/WR DATA 00
MEMAD/WDAT01	22	I	MEM ADR/WR DATA 01
MEMAD/WDAT02	23	I	MEM ADR/WR DATA 02
MEMAD/WDAT03	24	I	MEM ADR/WR DATA 03
RMW	26	I	RMW
DTR HLD OFF	27	I	DTR HOLD OFF
MEMAD/WDAT04	28	I	MEM ADR/WR DATA 04
MEMAD/WDAT05	29	I	MEM ADR/WR DATA 05
MEMAD/WDAT06	30	I	MEM ADR/WR DATA 06
MEMAD/WDAT07	32	I	MEM ADR/WR DATA 07
CB ABRT	34	I	CB ABORT
CB MI LPBK	35	I	CB MI LOOPBACK
MEM BSY	36	I	MEM BUSY
CB FCTN 0	38	I	CB FCTN 0
CB FCTN 1	39	I	CB FCTN 1
CB FCTN 2	41	I	CB FCTN 2
CB FCTN 3	42	I	CB FCTN 3
LTCH FCTN	43	I	LATCH FCTN
CLR/TMOUT	46	I	CLEAR/TIMEOUT

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6.4.19 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CORE	47	I	CORE
INTRNL ATR	50	I	INTERNAL ATR
MEM PAR ERR	52	I	MEM PARITY ERROR
WT RAM	72	I	WR RAM
MMIO CYC	73	I	MMIO CYCLE
MEMAD/WDAT08	76	I	MEM ADR/WR DATA 08
MEMAD/WDAT09	77	I	MEM ADR/WR DATA 09
MEMAD/WDAT10	78	I	MEM ADR/WR DATA 10
MEMAD/WDAT11	79	I	MEM ADR/WR DATA 11
MEMAD/WDAT12	80	I	MEM ADR/WR DATA 12
MEMAD/WDAT13	82	I	MEM ADR/WR DATA 13
MEMAD/WDAT14	83	I	MEM ADR/WR DATA 14
MEMAD/WDAT15	84	I	MEM ADR/WR DATA 15
MODULE CD A	85	I	MODULE CODE A
MODULE EN 0	86	I	MODULE ENBL 0
MODULE EN 1	87	I	MODULE ENBL 1
MODULE EN 2	88	I	MODULE ENBL 2
MODULE EN 3	89	I	MODULE ENBL 3
MODULE CD B	95	I	MODULE CODE B
MODULE CD C	96	I	MODULE CODE C

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6.4.19 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM RD DAT00	11	I	MEM RD DATA 0
MEM RD DAT01	12	I	MEM RD DATA 1
MEM RD DAT02	13	I	MEM RD DATA 2
MEM RD DAT03	14	I	MEM RD DATA 3
MEM RD DAT04	15	I	MEM RD DATA 4
MEM RD DAT05	16	I	MEM RD DATA 5
MEM RD DAT06	17	I	MEM RD DATA 6
MEM RD DAT07	18	I	MEM RD DATA 7
DAT RDY	20	I	DATA READY
MEM RD DAT08	64	I	MEM RD DATA 8
MEM RD DAT09	65	I	MEM RD DATA 9
MEM RD DAT10	66	I	MEM RD DATA 10
MEM RD DAT11	67	I	MEM RD DATA 11
MEM RD DAT12	68	I	MEM RD DATA 12
MEM RD DAT13	69	I	MEM RD DATA 13
MEM RD DAT14	70	I	MEM RD DATA 14
MEM RD DAT15	71	I	MEM RD DATA 15
CB RD DAT07	2	O	CB RD DATA 7
CB RD DAT06	3	O	CB RD DATA 6
CB RD DAT05	4	O	CB RD DATA 5

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6.4.19 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB RD DAT04	5	O	CB RD DATA 4
CB RD DAT03	6	O	CB RD DATA 3
CB RD DAT02	7	O	CB RD DATA 2
CB RD DAT01	8	O	CB RD DATA 1
CB RD DAT00	9	O	CB RD DATA 0
CLR BUS BSY	33	O	CLR BUS BUSY
LTCH ABRT	37	O	LATCHED ABORT
ACC STS	44	O	ACCESS STATUS
2 WRD RD	45	O	TWO WORD READ
CB PAR ERR	53	O	CB PARITY ERROR
CB DTR	54	O	CB DTR
CB RD DAT08	55	O	CB RD DATA 8
CB RD DAT09	56	O	CB RD DATA 9
CB RD DAT10	57	O	CB RD DATA 10
CB RD DAT11	58	O	CB RD DATA 11
CB RD DAT12	59	O	CB RD DATA 12
CB RD DAT13	61	O	CB RD DATA 13
CB RD DAT14	62	O	CB RD DATA 14
CB RD DAT15	63	O	CB RD DATA 15
LPBX CYC	74	O	LOOPBACK CYCLE

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6.4.19 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
FCTN 0	91	O	FCTN 0
FCTN 1	92	O	FCTN 1
FCTN 2	93	O	FCTN 2
FCTN 3	94	O	FCTN 3
WT BYTE LOWR	97	O	WR BYTE LOWER
WT BYTE UPPR	98	O	WR BYTE UPPER
1ST HF 2WD	100	O	1ST HALF 2 WORD CORE

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N741)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C	Frame	10	INF	MOhm	1-8
A2	Isolation used to unused	C	Frame				
A3	Isolation unused to unused	48	49,99				
A4	Isolation unused to unused	49	99	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A4
Isolation measurement	NA	± 500 KOhm	A1-A4

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

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NOTES: - Continued.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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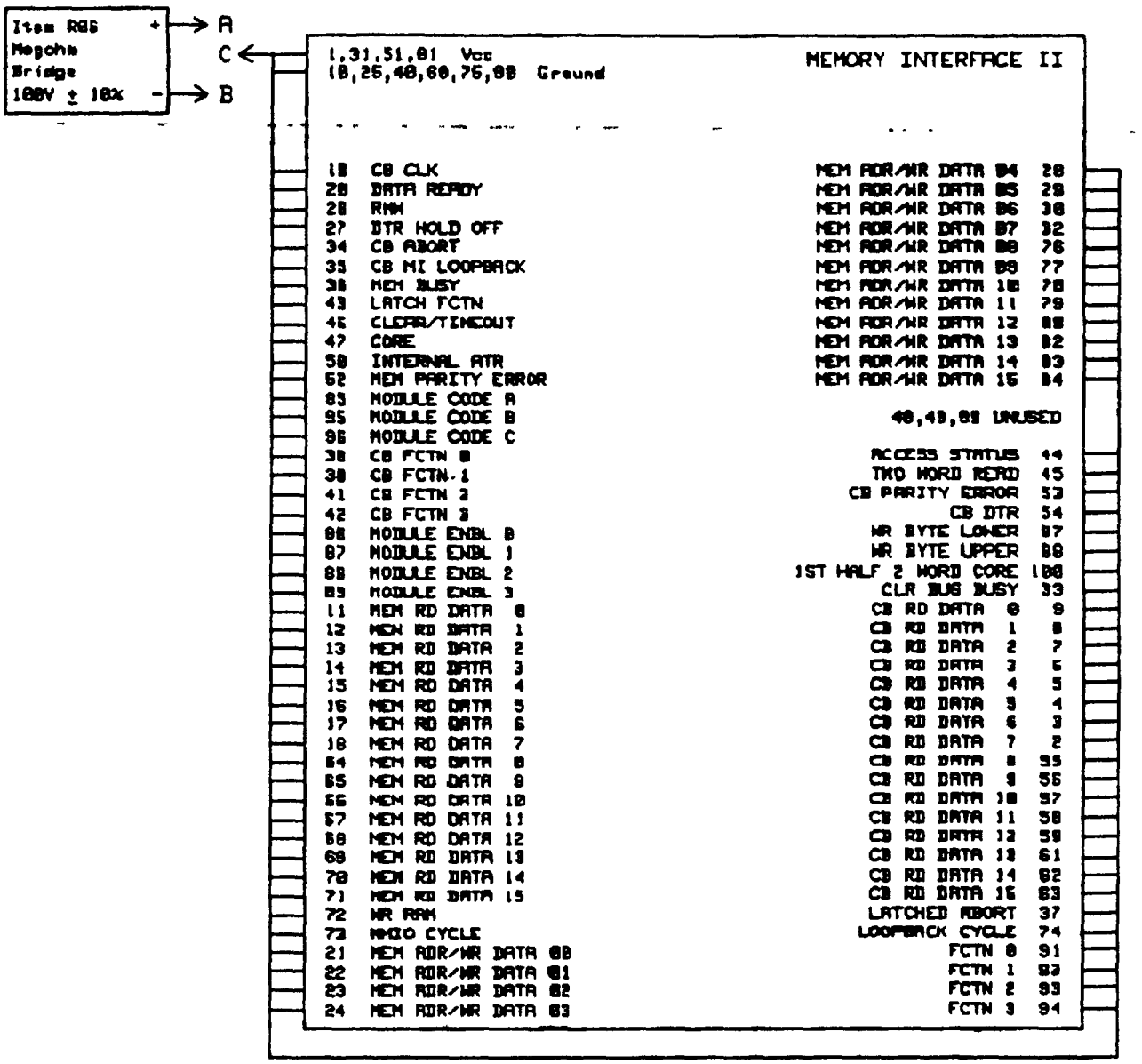


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

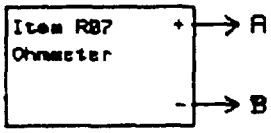
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 Vcc
10,25,40,60,75,90 Ground

MEMORY INTERFACE II

19	CB CLK	MEM ADR/WR DATA 04	28
20	DATA READY	MEM ADR/WR DATA 05	29
26	RMW	MEM ADR/WR DATA 06	30
27	DTR HOLD OFF	MEM ADR/WR DATA 07	32
34	CB ABORT	MEM ADR/WR DATA 08	76
35	CB MI LOOPBACK	MEM ADR/WR DATA 09	77
36	MEM BUSY	MEM ADR/WR DATA 10	78
43	LATCH FCTN	MEM ADR/WR DATA 11	79
45	CLEAR/TIMEOUT	MEM ADR/WR DATA 12	80
47	CORE	MEM ADR/WR DATA 13	82
50	INTERNAL ATR	MEM ADR/WR DATA 14	83
52	MEM PARITY ERROR	MEM ADR/WR DATA 15	84
85	MODULE CODE A		
95	MODULE CODE B	40,49,89 UNUSED	
96	MODULE CODE C		
38	CB FCTN 0	ACCESS STATUS	44
39	CB FCTN 1	TWO WORD READ	45
41	CB FCTN 2	CB PARITY ERROR	53
42	CB FCTN 3	CB DTR	54
86	MODULE ENBL 0	WR BYTE LOWER	97
87	MODULE ENBL 1	WR BYTE UPPER	98
88	MODULE ENBL 2	1ST HALF 2 WORD CORE	100
89	MODULE ENBL 3	CLR BUS BUSY	33
11	MEM RD DATA 0	CB RD DATA 0	9
12	MEM RD DATA 1	CB RD DATA 1	8
13	MEM RD DATA 2	CB RD DATA 2	7
14	MEM RD DATA 3	CB RD DATA 3	6
15	MEM RD DATA 4	CB RD DATA 4	5
16	MEM RD DATA 5	CB RD DATA 5	4
17	MEM RD DATA 6	CB RD DATA 6	3
18	MEM RD DATA 7	CB RD DATA 7	2
64	MEM RD DATA 8	CB RD DATA 8	55
65	MEM RD DATA 9	CB RD DATA 9	56
66	MEM RD DATA 10	CB RD DATA 10	57
67	MEM RD DATA 11	CB RD DATA 11	58
68	MEM RD DATA 12	CB RD DATA 12	59
69	MEM RD DATA 13	CB RD DATA 13	61
70	MEM RD DATA 14	CB RD DATA 14	62
71	MEM RD DATA 15	CB RD DATA 15	63
72	WR RAM	LATCHED ABORT	97
73	MMIO CYCLE	LOOPBACK CYCLE	74
21	MEM ADR/WR DATA 00	FCTN 0	91
22	MEM ADR/WR DATA 01	FCTN 1	92
23	MEM ADR/WR DATA 02	FCTN 2	93
24	MEM ADR/WR DATA 03	FCTN 3	94

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.55 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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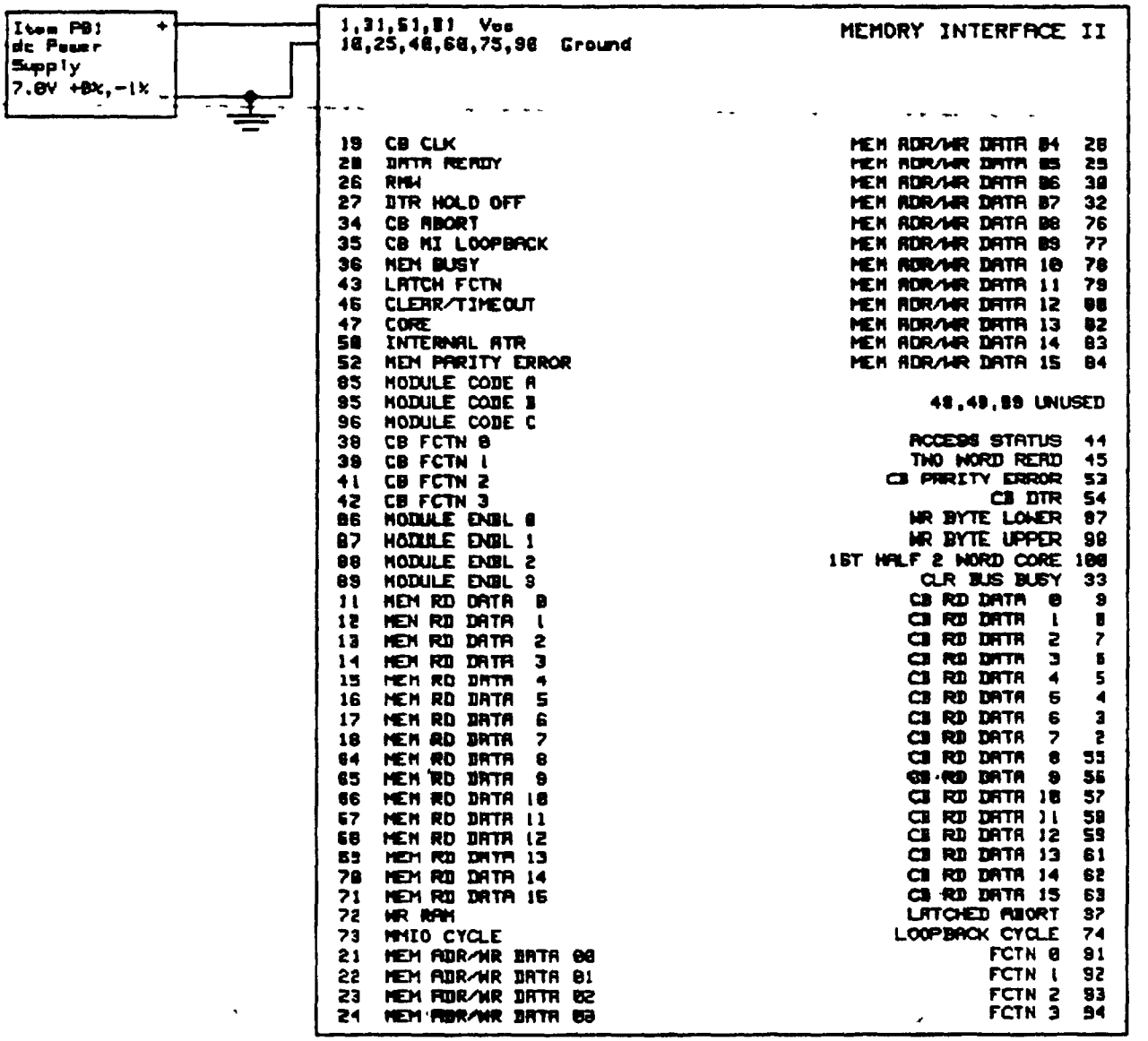


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CB CLK	19	5.5	0.0	1.0	mA	1-3
D2	MEMAD/WDAT00	21	5.5	0.0	0.2	mA	1-3
D3	MEMAD/WDAT01	22					
D4	MEMAD/WDAT02	23					
D5	MEMAD/WDAT03	24					
D6	RMW	26	5.5	0.0	0.2	mA	1-3
D7	DTR HLD OFF	27	5.5	0.0	0.3	mA	1-3
D8	MEMAD/WDAT04	28	5.5	0.0	0.2	mA	1-3
D9	MEMAD/WDAT05	29					
D10	MEMAD/WDAT06	30					
D11	MEMAD/WDAT07	32	5.5	0.0	0.2	mA	1-3
D12	CB ABRT	34	5.5	0.0	0.1	mA	1-3
D13	CB MI LPBK	35	5.5	0.0	0.1	mA	1-3
D14	MEM BSY	36	5.5	0.0	1.0	mA	1-3
D15	CB FCTN 0	38	5.5	0.0	0.1	mA	1-3
D16	CB FCTN 1	39					
D17	CB FCTN 2	41					
D18	CB FCTN 3	42					
D19	LTCH FCTN	43					
D20	CLR/TMOUT	46	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	CORE	47	5.5	0.0	0.3	mA	1-3
D22	INTRNL ATR	50	5.5	0.0	0.3	mA	1-3
D23	MEM PAR ERR	52	5.5	-0.12	0.22	mA	1-3
D24	WT RAM	72	5.5	0.0	0.1	mA	1-3
D25	MMIO CYC	73	5.5	0.0	0.2	mA	1-3
D26	MEMAD/WDAT08	76	5.5	0.0	0.1	mA	1-3
D27	MEMAD/WDAT09	77					
D28	MEMAD/WDAT10	78					
D29	MEMAD/WDAT11	79					
D30	MEMAD/WDAT12	80					
D31	MEMAD/WDAT13	82					
D32	MEMAD/WDAT14	83					
D33	MEMAD/WDAT15	84	5.5	0.0	0.1	mA	1-3
D34	MODULE CD A	85	5.5	-0.12	0.22	mA	1-3
D35	MODULE EN 0	86	5.5	0.0	0.1	mA	1-3
D36	MODULE EN 1	87					
D37	MODULE EN 2	88					
D38	MODULE EN 3	89	5.5	0.0	0.1	mA	1-3
D39	MODULE CD B	95	5.5	-0.12	0.22	mA	1-3
D40	MODULE CD C	96	5.5	-0.12	0.22	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D41	MEM RD DAT00	11	5.5	22.93	27.32	mA	1-3
D42	MEM RD DAT01	12					
D43	MEM RD DAT02	13					
D44	MEM RD DAT03	14					
D45	MEM RD DAT04	15					
D46	MEM RD DAT05	16					
D47	MEM RD DAT06	17					
D48	MEM RD DAT07	18	5.5	22.93	27.32	mA	1-3
D49	DAT RDY	20	5.5	22.93	29.32	mA	1-3
D50	MEM RD DAT08	64	5.5	22.93	27.32	mA	1-3
D51	MEM RD DAT09	65					
D52	MEM RD DAT10	66					
D53	MEM RD DAT11	67					
D54	MEM RD DAT12	68					
D55	MEM RD DAT13	69					
D56	MEM RD DAT14	70					
D57	MEM RD DAT15	71	5.5	22.93	27.32	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D57
Power supply current limit	2.12 A	NA	D1-D57
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D57
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D57 D1-D57
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 50 uA ± 10 uA ± 15 uA ± 10 uA ± 5 uA ± 50 uA ± 5 uA ± 15 uA ± 17 uA ± 5 uA ± 10 uA ± 5 uA ± 17 uA ± 5 uA ± 17 uA ± 219 uA ± 320 uA ± 219 uA	D1 D2-D6 D7 D8-D11 D12,D13 D14 D15-D20 D21,D22 D23 D24 D25 D26-D33 D34 D35-D38 D39,D40 D41-D48 D49 D50-D57

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CB RD DAT07	2	0.0	-225.0	-100.0	mA	1-4
E2	CB RD DAT06	3					
E3	CB RD DAT05	4					
E4	CB RD DAT04	5					
E5	CB RD DAT03	6					
E6	CB RD DAT02	7					
E7	CB RD DAT01	8					
E8	CB RD DAT00	9	0.0	-225.0	-100.0	mA	1-4
E9	CLR BUS BSY	33	0.0	-150.0	-60.0	mA	1-4
E10	LTCH ABRT	37					
E11	ACC STS	44					
E12	2 WRD RD	45	0.0	-150.0	-60.0	mA	1-4
E13	CB PAR ERR	53	0.0	-225.0	-100.0	mA	1-4
E14	CB DTR	54					
E15	CB RD DAT08	55					
E16	CB RD DAT09	56					
E17	CB RD DAT10	57					
E18	CB RD DAT11	58					
E19	CB RD DAT12	59					
E20	CB RD DAT13	61	0.0	-225.0	-100.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	CB RD DAT14	62	0.0	-225.0	-100.0	mA	1-4
E22	CB RD DAT15	63	0.0	-225.0	-100.0	mA	1-4
E23	LPBX CYC	74	0.0	-150.0	-60.0	mA	1-4
E24	FCTN 0	91	0.0	-100.0	-40.0	mA	1-4
E25	FCTN 1	92					
E26	FCTN 2	93					
E27	FCTN 3	94	0.0	-100.0	-40.0	mA	1-4
E28	WT BYTE LOWR	97	0.0	-150.0	-60.0	mA	1-4
E29	WT BYTE UPPR	98					
E30	1ST HF 2WD	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E30
Power supply current limit	2.12 A	NA	E1-E30
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E30
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E30 E1-E30
Output current measurement	NA NA NA NA NA NA	± 6.25 mA ± 4.5 mA ± 6.25 mA ± 4.5 mA ± 3.0 mA ± 4.5 mA	E1-E8 E9-E12 E13-E22 E23 E24-E27 E28-E30

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CB CLK	19	2.4	-2.0	0.05	mA	1-3
F2	MEMAD/WDAT00	21	2.4	-0.4	0.04	mA	1-3
F3	MEMAD/WDAT01	22					
F4	MEMAD/WDAT02	23					
F5	MEMAD/WDAT03	24	2.4	-0.4	0.04	mA	1-3
F6	RMW	26	2.4	-1.2	0.04	mA	1-3
F7	DTR HLD OFF	27	2.4	-1.8	0.06	mA	1-3
F8	MEMAD/WDAT04	28	2.4	-0.4	0.04	mA	1-3
F9	MEMAD/WDAT05	29					
F10	MEMAD/WDAT06	30					
F11	MEMAD/WDAT07	32	2.4	-0.4	0.04	mA	1-3
F12	CB ABRT	34	2.4	-0.6	0.02	mA	1-3
F13	CB MI LPBK	35	2.4	-0.6	0.02	mA	1-3
F14	MEM BSY	36	2.4	-2.0	0.05	mA	1-3
F15	CB FCTN 0	38	2.4	-0.6	0.02	mA	1-3
F16	CB FCTN 1	39					
F17	CB FCTN 2	41					
F18	CB FCTN 3	42					
F19	LTCH FCTN	43					
F20	CLR/TMOUT	46	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	CORE	47	2.4	-1.4	0.06	mA	1-3
F22	INTRNL ATR	50	2.4	-0.6	0.06	mA	1-3
F23	MEM PAR ERR	52	2.4	-7.23	-2.85	mA	1-3
F24	WT RAM	72	2.4	-0.2	0.02	mA	1-3
F25	MMIO CYC	73	2.4	-1.2	0.04	mA	1-3
F26	MEMAD/WDAT08	76	2.4	-0.2	0.02	mA	1-3
F27	MEMAD/WDAT09	77					
F28	MEMAD/WDAT10	78					
F29	MEMAD/WDAT11	79					
F30	MEMAD/WDAT12	80					
F31	MEMAD/WDAT13	82					
F32	MEMAD/WDAT14	83					
F33	MEMAD/WDAT15	84	2.4	-0.2	0.02	mA	1-3
F34	MODULE CD A	85	2.4	-5.63	-2.85	mA	1-3
F35	MODULE EN 0	86	2.4	-0.2	0.02	mA	1-3
F36	MODULE EN 1	87					
F37	MODULE EN 2	88					
F38	MODULE EN 3	89	2.4	-0.2	0.02	mA	1-3
F39	MODULE CD B	95	2.4	-5.63	-2.85	mA	1-3
F40	MODULE CD C	96	2.4	-5.63	-2.85	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	MEM RD DAT00	11	2.4	-28.26	-4.36	mA	1-3
F42	MEM RD DAT01	12					
F43	MEM RD DAT02	13					
F44	MEM RD DAT03	14					
F45	MEM RD DAT04	15					
F46	MEM RD DAT05	16					
F47	MEM RD DAT06	17					
F48	MEM RD DAT07	18	2.4	-28.26	-4.36	mA	1-3
F49	DAT RDY	20	2.4	-35.06	-4.21	mA	1-3
F50	MEM RD DAT08	64	2.4	-28.26	-4.36	mA	1-3
F51	MEM RD DAT09	65					
F52	MEM RD DAT10	66					
F53	MEM RD DAT11	67					
F54	MEM RD DAT12	68					
F55	MEM RD DAT13	69					
F56	MEM RD DAT14	70					
F57	MEM RD DAT15	71	2.4	-28.26	-4.36	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F57
Power supply current limit	2.12 A	NA	F1-F57
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F57
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F57
low, (VL)	0.0 V	± 100 mV	F1-F57
Input current measurement	NA	± 102 uA	F1
	NA	± 22 uA	F2-F5
	NA	± 62 uA	F6
	NA	± 93 uA	F7
	NA	± 22 uA	F8-F11
	NA	± 31 uA	F12,F13
	NA	± 102 uA	F14
	NA	± 31 uA	F15-F20
	NA	± 73 uA	F21
	NA	± 33 uA	F22
	NA	± 219 uA	F23
	NA	± 11 uA	F24
	NA	± 62 uA	F25
	NA	± 11 uA	F26-F33
	NA	± 139 uA	F34
	NA	± 11 uA	F35-F38
	NA	± 139 uA	F39,F40
	NA	± 1190 uA	F41-F48
	NA	± 1540 uA	F49
	NA	± 1190 uA	F50-F57

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CB CLK	19	0.4	-2.0	0.05	mA	1-3
G2	MEMAD/WDAT00	21	0.4	-0.4	0.04	mA	1-3
G3	MEMAD/WDAT01	22					
G4	MEMAD/WDAT02	23					
G5	MEMAD/WDAT03	24	0.4	-0.4	0.04	mA	1-3
G6	RMW	26	0.4	-1.2	0.04	mA	1-3
G7	DTR HLD OFF	27	0.4	-3.8	0.11	mA	1-3
G8	MEMAD/WDAT04	28	0.4	-0.4	0.04	mA	1-3
G9	MEMAD/WDAT05	29					
G10	MEMAD/WDAT06	30					
G11	MEMAD/WDAT07	32	0.4	-0.4	0.04	mA	1-3
G12	CB ABRT	34	0.4	-0.6	0.02	mA	1-3
G13	CB MI LPBK	35	0.4	-0.6	0.02	mA	1-3
G14	MEM BSY	36	0.4	-2.0	0.05	mA	1-3
G15	CB FCTN 0	38	0.4	-0.6	0.02	mA	1-3
G16	CB FCTN 1	39					
G17	CB FCTN 2	41					
G18	CB FCTN 3	42					
G19	LTCH FCTN	43					
G20	CLR/TMOUT	46	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	CORE	47	0.4	-1.4	0.06	mA	1-3
G22	INTRNL ATR	50	0.4	-2.6	0.11	mA	1-3
G23	MEM PAR ERR	52	0.4	-7.23	-2.85	mA	1-3
G24	WT RAM	72	0.4	-0.2	0.02	mA	1-3
G25	MMIO CYC	73	0.4	-1.2	0.04	mA	1-3
G26	MEMAD/WDAT08	76	0.4	-0.2	0.02	mA	1-3
G27	MEMAD/WDAT09	77					
G28	MEMAD/WDAT10	78					
G29	MEMAD/WDAT11	79					
G30	MEMAD/WDAT12	80					
G31	MEMAD/WDAT13	82					
G32	MEMAD/WDAT14	83					
G33	MEMAD/WDAT15	84	0.4	-0.2	0.02	mA	1-3
G34	MODULE CD A	85	0.4	-5.63	-2.85	mA	1-3
G35	MODULE EN 0	86	0.4	-0.2	0.02	mA	1-3
G36	MODULE EN 1	87					
G37	MODULE EN 2	88					
G38	MODULE EN 3	89	0.4	-0.2	0.02	mA	1-3
G39	MODULE CD B	95	0.4	-5.63	-2.85	mA	1-3
G40	MODULE CD C	96	0.4	-5.63	-2.85	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	MEM RD DAT00	11	0.4	-28.26	-4.36	mA	1-3
G42	MEM RD DAT01	12					
G43	MEM RD DAT02	13					
G44	MEM RD DAT03	14					
G45	MEM RD DAT04	15					
G46	MEM RD DAT05	16					
G47	MEM RD DAT06	17					
G48	MEM RD DAT07	18	0.4	-28.26	-4.36	mA	1-3
G49	DAT RDY	20	0.4	-35.06	-4.21	mA	1-3
G50	MEM RD DAT08	64	0.4	-28.26	-4.36	mA	1-3
G51	MEM RD DAT09	65					
G52	MEM RD DAT10	66					
G53	MEM RD DAT11	67					
G54	MEM RD DAT12	68					
G55	MEM RD DAT13	69					
G56	MEM RD DAT14	70					
G57	MEM RD DAT15	71	0.4	-28.26	-4.36	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G57
Power supply current limit	2.12 A	NA	G1-G57
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G57
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G57
low, (VL)	0.0 V	± 100 mV	G1-G57
Input current measurement	NA	± 102 uA	G1
	NA	± 22 uA	G2-G5
	NA	± 62 uA	G6
	NA	± 195 uA	G7
	NA	± 22 uA	G8-G11
	NA	± 31 uA	G12,G13
	NA	± 102 uA	G14
	NA	± 31 uA	G15-G20
	NA	± 73 uA	G21
	NA	± 135 uA	G22
	NA	± 219 uA	G23
	NA	± 11 uA	G24
	NA	± 62 uA	G25
	NA	± 11 uA	G26-G33
	NA	± 139 uA	G34
	NA	± 11 uA	G35-G38
	NA	± 139 uA	G39,G40
	NA	± 1190 uA	G41-G48
	NA	± 1540 uA	G49
	NA	± 1190 uA	G50-G57

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	CB RD DAT07	2	-0.4	2.4	4.5	V	1-4
H2	CB RD DAT06	3					
H3	CB RD DAT05	4					
H4	CB RD DAT04	5					
H5	CB RD DAT03	6					
H6	CB RD DAT02	7					
H7	CB RD DAT01	8					
H8	CB RD DAT00	9					
H9	CLR BUS BSY	33					
H10	LTCH ABRT	37					
H11	ACC STS	44					
H12	2 WRD RD	45					
H13	CB PAR ERR	53					
H14	CB DTR	54					
H15	CB RD DAT08	55					
H16	CB RD DAT09	56					
H17	CB RD DAT10	57					
H18	CB RD DAT11	58					
H19	CB RD DAT12	59					
H20	CB RD DAT13	61	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	CB RD DAT14	62	-0.4	2.4	4.5	V	1-4
H22	CB RD DAT15	63					
H23	LPBX CYC	74					
H24	FCTN 0	91					
H25	FCTN 1	92					
H26	FCTN 2	93					
H27	FCTN 3	94					
H28	WT BYTE LOWR	97					
H29	WT BYTE UPPR	98					
H30	1ST HF 2WD	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H30
Power supply current limit	2.12 A	NA	H1-H30
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H30
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H30 H1-H30
Output voltage measurement	NA	± 105 mV	H1-H30

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

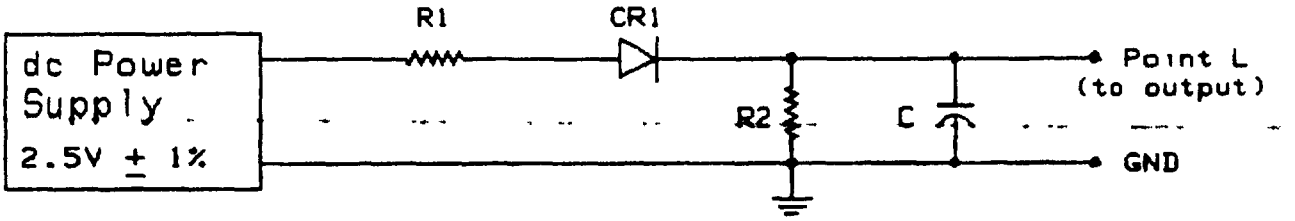
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 33,37,44,45,53,54, 55,56,57,58,59,61, 62,63,74,91,92,93, 94,97,98,100	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	CB RD DAT07	2	6.0	0.0	0.5	V	1-4
J2	CB RD DAT06	3					
J3	CB RD DAT05	4					
J4	CB RD DAT04	5					
J5	CB RD DAT03	6					
J6	CB RD DAT02	7					
J7	CB RD DAT01	8					
J8	CB RD DAT00	9					
J9	CLR BUS BSY	33					
J10	LTCH ABRT	37					
J11	ACC STS	44					
J12	2 WRD RD	45					
J13	CB PAR ERR	53					
J14	CB DTR	54					
J15	CB RD DAT08	55					
J16	CB RD DAT09	56					
J17	CB RD DAT10	57					
J18	CB RD DAT11	58					
J19	CB RD DAT12	59					
J20	CB RD DAT13	61	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	CB RD DAT14	62	6.0	0.0	0.5	V	1-4
J22	CB RD DAT15	63					
J23	LPBX CYC	74					
J24	FCTN 0	91					
J25	FCTN 1	92					
J26	FCTN 2	93					
J27	FCTN 3	94					
J28	WT BYTE LOWR	97					
J29	WT BYTE UPPR	98					
J30	1ST HF 2WD	100	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:39Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J30
Power supply current limit	2.12 A	NA	J1-J30
Forcing current, (Iol)	6.0 uA	± 600 uA	J1-J30
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J30 J1-J30
Output voltage measurement	NA	± 25 mV	J1-J30

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	CB RD DAT07	2	2.4	-0.05	0.05	mA	1-3
K2	CB RD DAT06	3					
K3	CB RD DAT05	4					
K4	CB RD DAT04	5					
K5	CB RD DAT03	6					
K6	CB RD DAT02	7					
K7	CB RD DAT01	8					
K8	CB RD DAT00	9					
K9	CB PAR ERR	53					
K10	CB DTR	54					
K11	CB RD DAT08	55					
K12	CB RD DAT09	56					
K13	CB RD DAT10	57					
K14	CB RD DAT11	58					
K15	CB RD DAT12	59					
K16	CB RD DAT13	61					
K17	CB RD DAT14	62					
K18	CB RD DAT15	63	2.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
Low level							
K19	CB RD DAT07	2	0.4	-0.05	0.05	mA	1-3
K20	CB RD DAT06	3					
K21	CB RD DAT05	4					
K22	CB RD DAT04	5					
K23	CB RD DAT03	6					
K24	CB RD DAT02	7					
K25	CB RD DAT01	8					
K26	CB RD DAT00	9					
K27	CB PAR ERR	53					
K28	CB DTR	54					
K29	CB RD DAT08	55					
K30	CB RD DAT09	56					
K31	CB RD DAT10	57					
K32	CB RD DAT11	58					
K33	CB RD DAT12	59					
K34	CB RD DAT13	61					
K35	CB RD DAT14	62					
K36	CB RD DAT15	63	0.4	-0.05	0.05	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:39Z
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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K36
Power supply current limit	2.12 A	NA	K1-K36
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K18 K19-K36
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K36 K1-K36
Output current measurement	NA	± 5 uA	K1-K36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CB CLK	19	-18.0	-1.2	0.0	V	1-3
L2	MEMAD/WDAT00	21	-18.0	-1.5	0.0	V	1-3
L3	MEMAD/WDAT01	22					
L4	MEMAD/WDAT02	23					
L5	MEMAD/WDAT03	24	-18.0	-1.5	0.0	V	1-3
L6	RMW	26	-18.0	-1.2	0.0	V	1-3
L7	DTR HLD OFF	27	-18.0	-1.2	0.0	V	1-3
L8	MEMAD/WDAT04	28	-18.0	-1.5	0.0	V	1-3
L9	MEMAD/WDAT05	29					
L10	MEMAD/WDAT06	30					
L11	MEMAD/WDAT07	32	-18.0	-1.5	0.0	V	1-3
L12	CB ABRT	34	-18.0	-1.2	0.0	V	1-3
L13	CB MI LPBK	35					
L14	MEM BSY	36					
L15	CB FCTN 0	38					
L16	CB FCTN 1	39					
L17	CB FCTN 2	41					
L18	CB FCTN 3	42					
L19	LTCH FCTN	43					
L20	CLR/TMOUT	46	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	CORE	47	-18.0	-1.2	0.0	V	1-3
L22	INTRNL ATR	50	-18.0	-1.5	0.0	V	1-3
L23	WT RAM	72	-18.0	-1.5	0.0	V	1-3
L24	MMIO CYC	73	-18.0	-1.2	0.0	V	1-3
L25	MEMAD/WDAT08	76	-18.0	-1.5	0.0	V	1-3
L26	MEMAD/WDAT09	77					
L27	MEMAD/WDAT10	78					
L28	MEMAD/WDAT11	79					
L29	MEMAD/WDAT12	80					
L30	MEMAD/WDAT13	82					
L31	MEMAD/WDAT14	83					
L32	MEMAD/WDAT15	84					
L33	MODULE EN 0	86					
L34	MODULE EN 1	87					
L35	MODULE EN 2	88					
L36	MODULE EN 3	89	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L36
Power supply current limit	NA	NA	L1-L36
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L36
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L36 L1-L36
Input voltage measurement	NA NA NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1 L2-L5 L6,L7 L8-L11 L12-L21 L22,L23 L24 L25-L36

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.6	6.8	uF	1-4
M2	PSFC	51	4.6	8.7	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

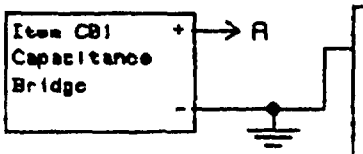
Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 160 nF ± 205 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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1,31,51,81 Vcc
10,25,40,60,75,90 Ground

MEMORY INTERFACE II

19	CB CLK	MEM ADR/WR DATA 04	20
20	DATA READY	MEM ADR/WR DATA 05	29
26	RMW	MEM ADR/WR DATA 06	30
27	DTR HOLD OFF	MEM ADR/WR DATA 07	32
34	CB ABORT	MEM ADR/WR DATA 08	76
35	CB MI LOOPBACK	MEM ADR/WR DATA 09	77
36	MEM BUSY	MEM ADR/WR DATA 10	78
43	LATCH FCTN	MEM ADR/WR DATA 11	79
46	CLEAR/TIMEOUT	MEM ADR/WR DATA 12	80
47	CORE	MEM ADR/WR DATA 13	82
50	INTERNAL ATR	MEM ADR/WR DATA 14	83
52	MEM PARITY ERROR	MEM ADR/WR DATA 15	84
85	MODULE CODE A		
95	MODULE CODE B	40,49,89 UNUSED	
96	MODULE CODE C		
38	CB FCTN 0	ACCESS STATUS	44
39	CB FCTN 1	TWO WORD READ	45
41	CB FCTN 2	CB PARITY ERROR	53
42	CB FCTN 3	CB DTR	54
86	MODULE ENBL 0	WR BYTE LOWER	87
87	MODULE ENBL 1	WR BYTE UPPER	98
88	MODULE ENBL 2	1ST HALF 2 WORD CORE	100
89	MODULE ENBL 3	CLR BUS BUSY	33
11	MEM RD DATA 0	CB RD DATA 0	9
12	MEM RD DATA 1	CB RD DATA 1	8
13	MEM RD DATA 2	CB RD DATA 2	7
14	MEM RD DATA 3	CB RD DATA 3	6
15	MEM RD DATA 4	CB RD DATA 4	5
16	MEM RD DATA 5	CB RD DATA 5	4
17	MEM RD DATA 6	CB RD DATA 6	3
18	MEM RD DATA 7	CB RD DATA 7	2
64	MEM RD DATA 8	CB RD DATA 8	55
65	MEM RD DATA 9	CB RD DATA 9	56
66	MEM RD DATA 10	CB RD DATA 10	57
67	MEM RD DATA 11	CB RD DATA 11	58
68	MEM RD DATA 12	CB RD DATA 12	59
69	MEM RD DATA 13	CB RD DATA 13	61
70	MEM RD DATA 14	CB RD DATA 14	62
71	MEM RD DATA 15	CB RD DATA 15	63
72	WR RAM	LATCHED ABORT	37
73	MMIO CYCLE	LOOPBACK CYCLE	74
21	MEM ADR/WR DATA 00	FCTN 0	91
22	MEM ADR/WR DATA 01	FCTN 1	92
23	MEM ADR/WR DATA 02	FCTN 2	93
24	MEM ADR/WR DATA 03	FCTN 3	94

FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	590	1416	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.12 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 41.3 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1,P2 P1,P2
Power supply current limit	2.12 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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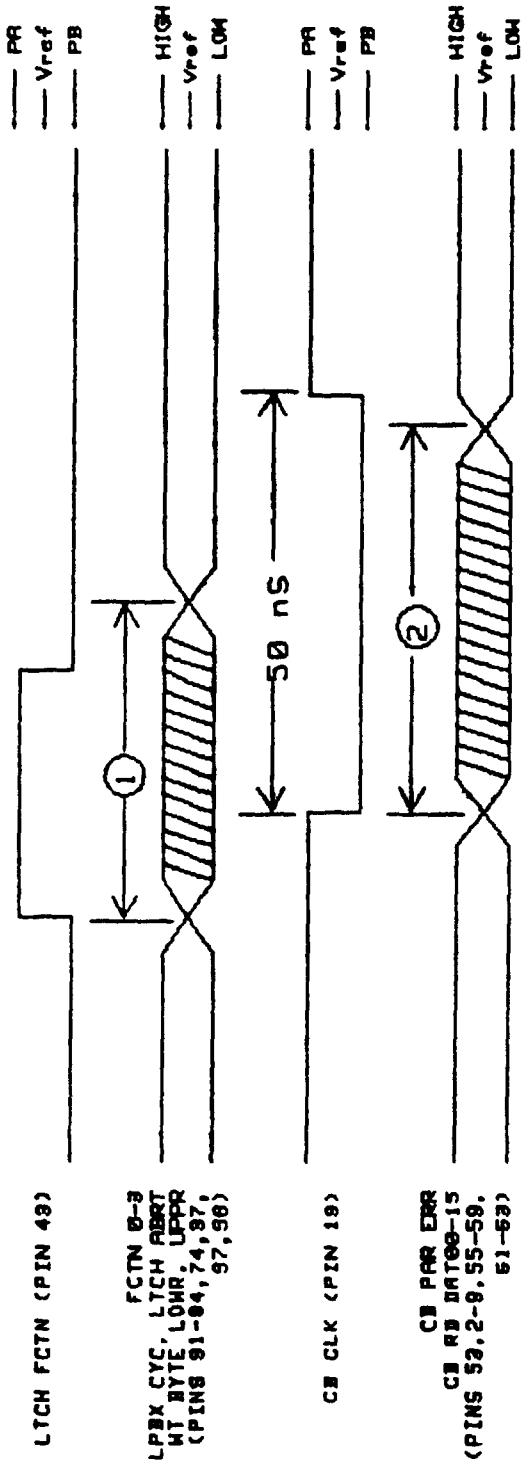


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (3) MEMAD/WDAT 00 - MEMAD/WDAT 07 to WT RAM setup time minimum of 30 nS.

2. (4) WT RAM to MEMAD/WDAT 00 - MEMAD/WDAT 07 hold time minimum of 4 nS.

3. (5) MODULE EN 0 - MODULE EN 3 and MEMAD/WDAT 08 - MEMAD/WDAT 15 to INTRNL ATR setup time minimum of 15 nS.

4. (6) INTRNL ATR to MODULE EN 0 - MODULE EN 3 and MEMAD WDAT 08 - MEMAD/WDAT 15 hold time minimum of 9 nS.

5. (7) MEM RD DAT 00 - MEM RD DAT 15 to DAT RDY setup time minimum of 15 nS.

6. (8) DAT RDY to MEM RD DAT 00 - MEM RD DAT 15 hold time minimum of 9 nS.

7. (9) MODULE CD A, MODULE CD B, MODULE CD C to MEM BSY setup time minimum of 15 nS.

8. (10) MEM BSY to MODULE CD A, MODULE CD B, MODULE CD C hold time minimum of 9 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.12 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:39Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

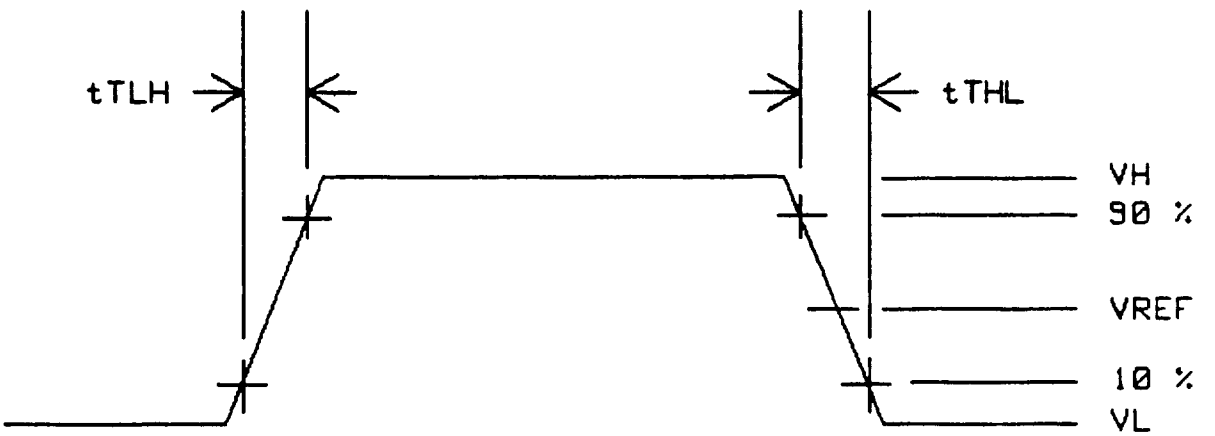
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	LTCH FCTN to FCTN 0	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to FCTN 1	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to FCTN 2	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to FCTN 3	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to LPBX FCTN	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to LTCH ABRT	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to WT BYTE LOWR	-	-	-	30	30	30	nS	12
1	tpd	LTCH FCTN to WT BYTE UPPR	-	-	-	30	30	30	nS	12
2	tpd	CB CLK to CB PAR ERR	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT07	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT06	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT05	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT04	-	-	-	40	40	40	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	CB CLK to CB RD DAT03	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT02	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT01	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT00	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT08	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT09	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT10	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT11	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT12	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT13	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT14	-	-	-	40	40	40	nS	12
2	tpd	CB CLK to CB RD DAT15	-	-	-	40	40	40	nS	12
3	tsu	MEMAD/WDAT00 to WT RAM	30	30	30	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tsu	MEMAD/WDAT01 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT02 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT03 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT04 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT05 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT06 to WT RAM	30	30	30	-	-	-	nS	12
3	tsu	MEMAD/WDAT07 to WT RAM	30	30	30	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT00	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT01	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT02	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT03	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT04	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT05	4	4	4	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	th	WT RAM to MEMAD/WDAT06	4	4	4	-	-	-	nS	12
4	th	WT RAM to MEMAD/WDAT07	4	4	4	-	-	-	nS	12
5	tsu	MODULE EN 0 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MODULE EN 1 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MODULE EN 2 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MODULE EN 3 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT08 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT09 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT10 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT11 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT12 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT13 to INTRNL ATR	15	15	15	-	-	-	nS	12
5	tsu	MEMAD/WDAT14 to INTRNL ATR	15	15	15	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	MEMAD/WDAT15 to INTRNL ATR	15	15	15	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 0	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 1	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 2	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MODULE EN 3	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT08	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT09	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT10	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT11	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT12	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT13	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT14	9	9	9	-	-	-	nS	12
6	th	INTRNL ATR to MEMAD/WDAT15	9	9	9	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEM RD DAT00 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT01 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT02 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT03 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT04 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT05 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT06 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT07 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT08 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT09 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT10 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT11 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT12 to DAT RDY	15	15	15	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEM RD DAT13 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT14 to DAT RDY	15	15	15	-	-	-	nS	12
7	tsu	MEM RD DAT15 to DAT RDY	15	15	15	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT00	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT01	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT02	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT03	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT04	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT05	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT06	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT07	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT08	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT09	9	9	9	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	DAT RDY to MEM RD DAT10	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT11	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT12	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT13	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT14	9	9	9	-	-	-	nS	12
8	th	DAT RDY to MEM RD DAT15	9	9	9	-	-	-	nS	12
9	tsu	MODULE CD A to MEM BSY	15	15	15	-	-	-	nS	12
9	tsu	MODULE CD B to MEM BSY	15	15	15	-	-	-	nS	12
9	tsu	MODULE CD C to MEM BSY	15	15	15	-	-	-	nS	12
10	th	MEM BSY to MODULE EN A	9	9	9	-	-	-	nS	12
10	th	MEM BSY to MODULE EN B	9	9	9	-	-	-	nS	12
10	th	MEM BSY to MODULE EN C	9	9	9	-	-	-	nS	12

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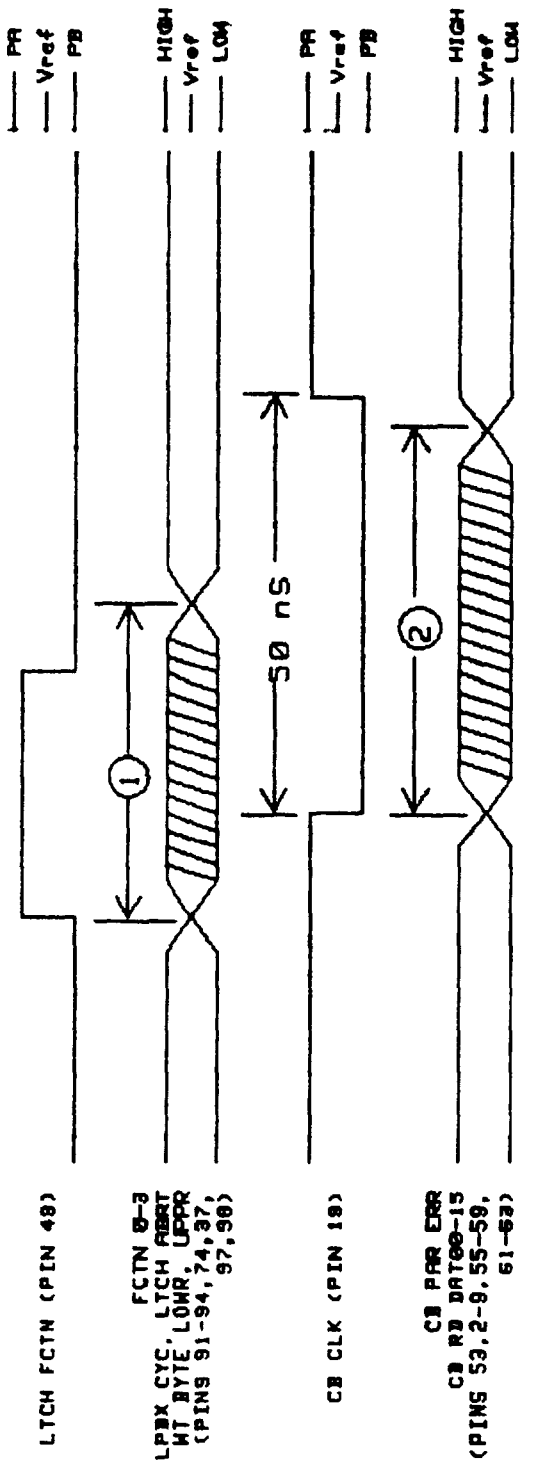


FIGURE 12a. Dynamic waveforms for table XVII.

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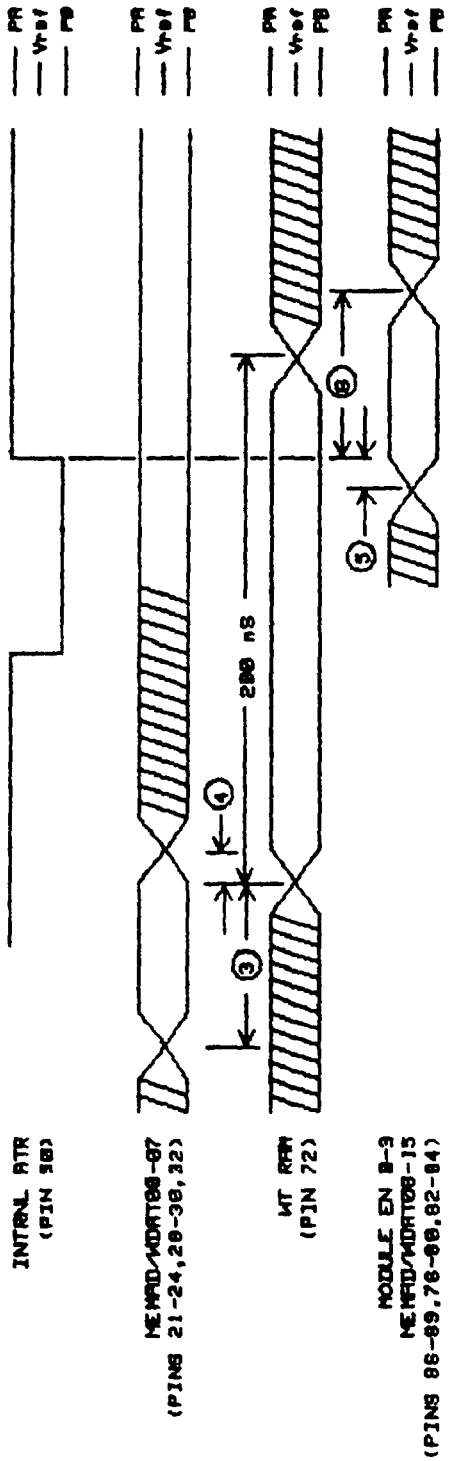


FIGURE 12b. Dynamic waveforms for table XVII.

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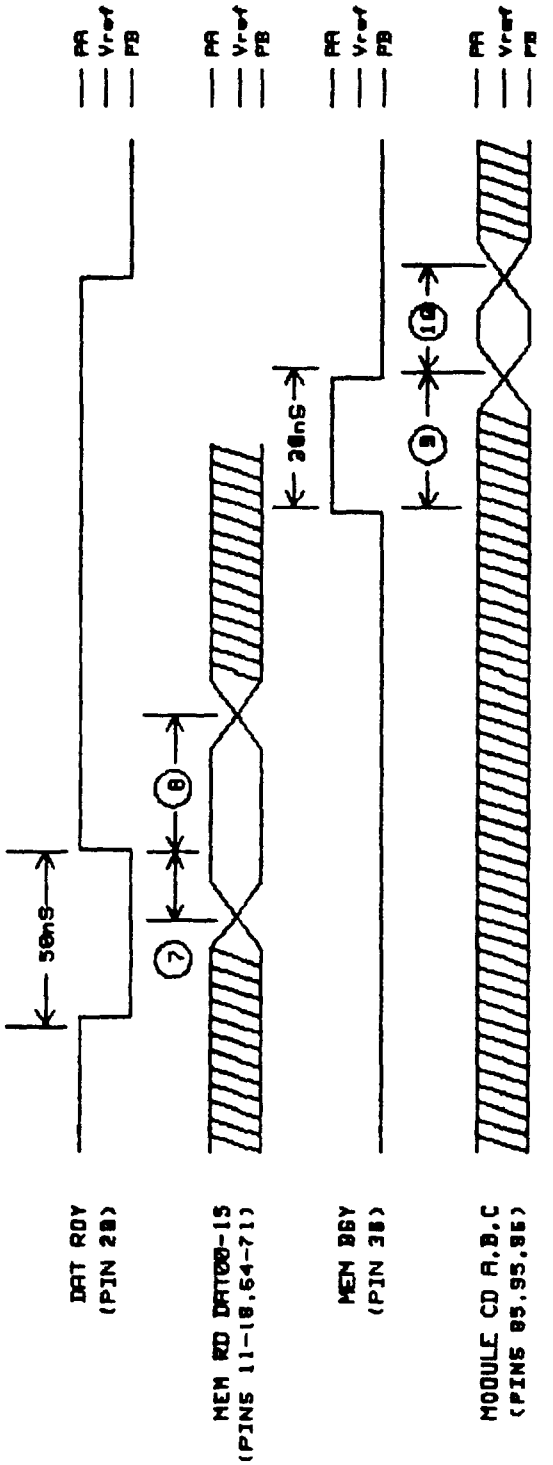


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	CB FCTN 0	38	2	50	pF	1-4
R2	CB FCTN 1	39				
R3	CB FCTN 2	41				
R4	CB FCTN 3	42				
R5	MODULE EN 0	86				
R6	MODULE EN 1	87				
R7	MODULE EN 2	88				
R8	MODULE EN 3	89	2	50	pF	1-4
R9	MODULE CD A	85	2	75	pF	1-4
R10	MODULE CD B	95	2	50	pF	1-4
R11	MODULE CD C	96	2	50	pF	1-4
R12	MEMAD/WDAT00	21	4	75	pF	1-4
R13	MEMAD/WDAT01	22	4	75	pF	1-4
R14	MEMAD/WDAT02	23	4	50	pF	1-4
R15	MEMAD/WDAT03	24	4	75	pF	1-4
R16	MEMAD/WDAT04	28				
R17	MEMAD/WDAT05	29				
R18	MEMAD/WDAT06	30				
R19	MEMAD/WDAT07	32	4	75	pF	1-4
R20	MEMAD/WDAT08	76	2	50	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	MEMAD/WDAT09	77	2	50	pF	1-4
R22	MEMAD/WDAT10	78				
R23	MEMAD/WDAT11	79				
R24	MEMAD/WDAT12	80				
R25	MEMAD/WDAT13	82				
R26	MEMAD/WDAT14	83				
R27	MEMAD/WDAT15	84				
R28	CB CLK	19	2	50	pF	1-4
R29	RMW	26	4	100	pF	1-4
R30	DTR HLD OFF	27	8	125	pF	1-4
R31	CB ABRT	34	2	50	pF	1-4
R32	CB MI LPBK	35				
R33	MEM BSY	36				
R34	LTCH FCTN	43	2	50	pF	1-4
R35	CLR/TMOUT	46	2	75	pF	1-4
R36	CORE	47	6	75	pF	1-4
R37	INTRNL ATR	50	8	150	pF	1-4
R38	MEM PAR ERR	52	2	50	pF	1-4
R39	WT RAM	72	2	75	pF	1-4
R40	MMIO CYC	73	4	100	pF	1-4

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R40
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R40
Capacitance measurement	NA	± 2.4 pF	R1-R8
	NA	± 3.65 pF	R9
	NA	± 2.4 pF	R10,R11
	NA	± 3.55 pF	R12,R13
	NA	± 2.3 pF	R14
	NA	± 3.55 pF	R15-R19
	NA	± 2.4 pF	R20-R28
	NA	± 4.8 pF	R29
	NA	± 5.85 pF	R30
	NA	± 2.4 pF	R31-R34
	NA	± 3.65 pF	R35
	NA	± 3.45 pF	R36
	NA	± 7.1 pF	R37
	NA	± 2.4 pF	R38
	NA	± 3.65 pF	R39
	NA	± 4.8 pF	R40

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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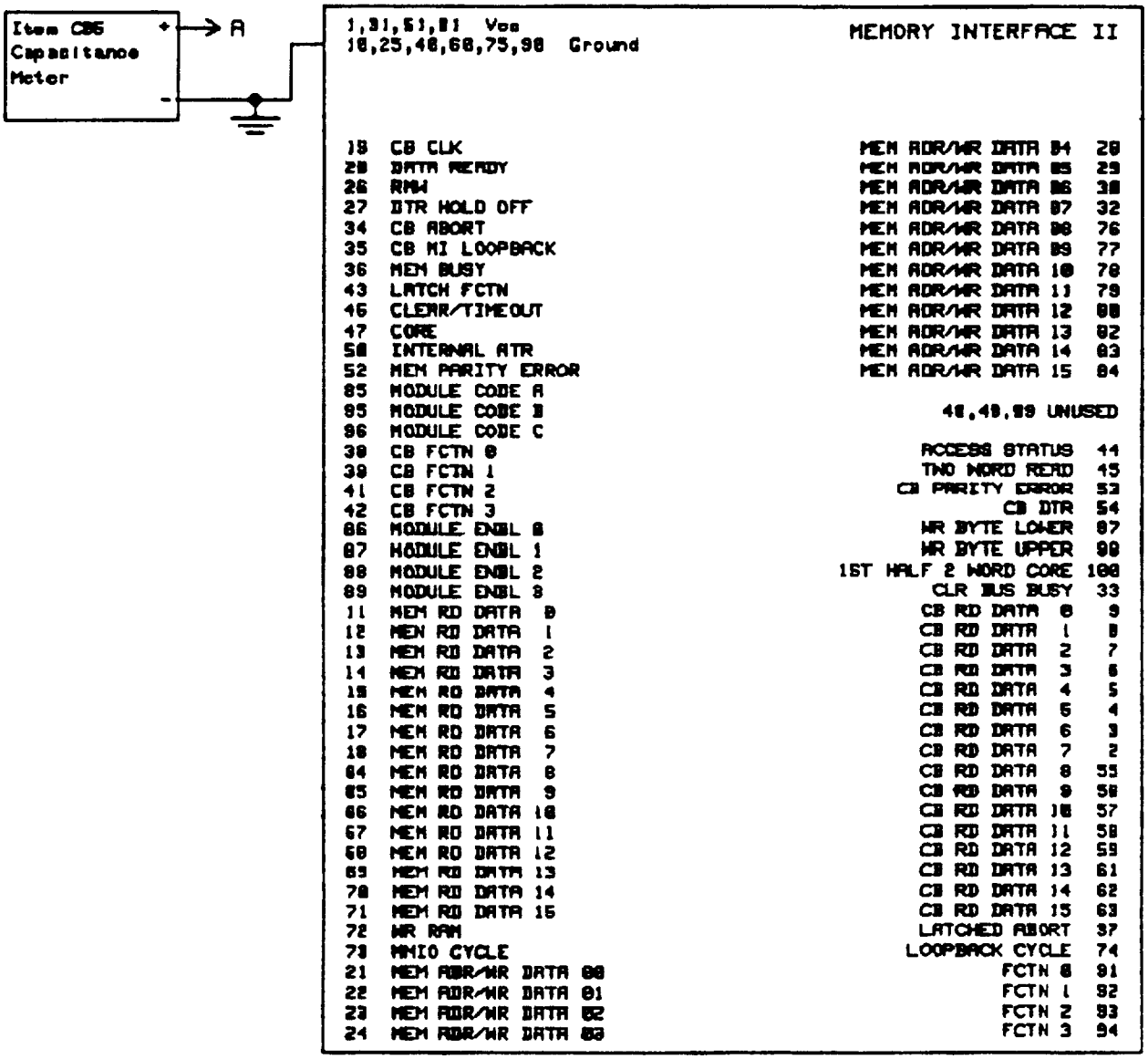


FIGURE 13. Input capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latchup test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S	LTCH FCTN	43	-18.0	-1.2	2.4	V	1-4

NOTES:

1. Input clamp diode latchup test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	S
Power supply current limit	2.12 A	NA	S
Forcing current, (Iin)	-18 mA	± 1.8 mA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Input voltage measurement	NA	± 180 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	450	mV	1-4
T2	PSTA	51	0	650	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.12 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	 NA NA	 ± 22.5 mV ± 32.5 mV	 T1 T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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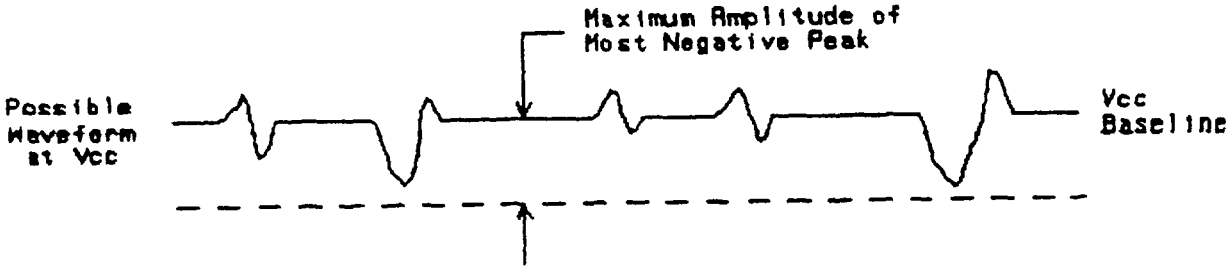
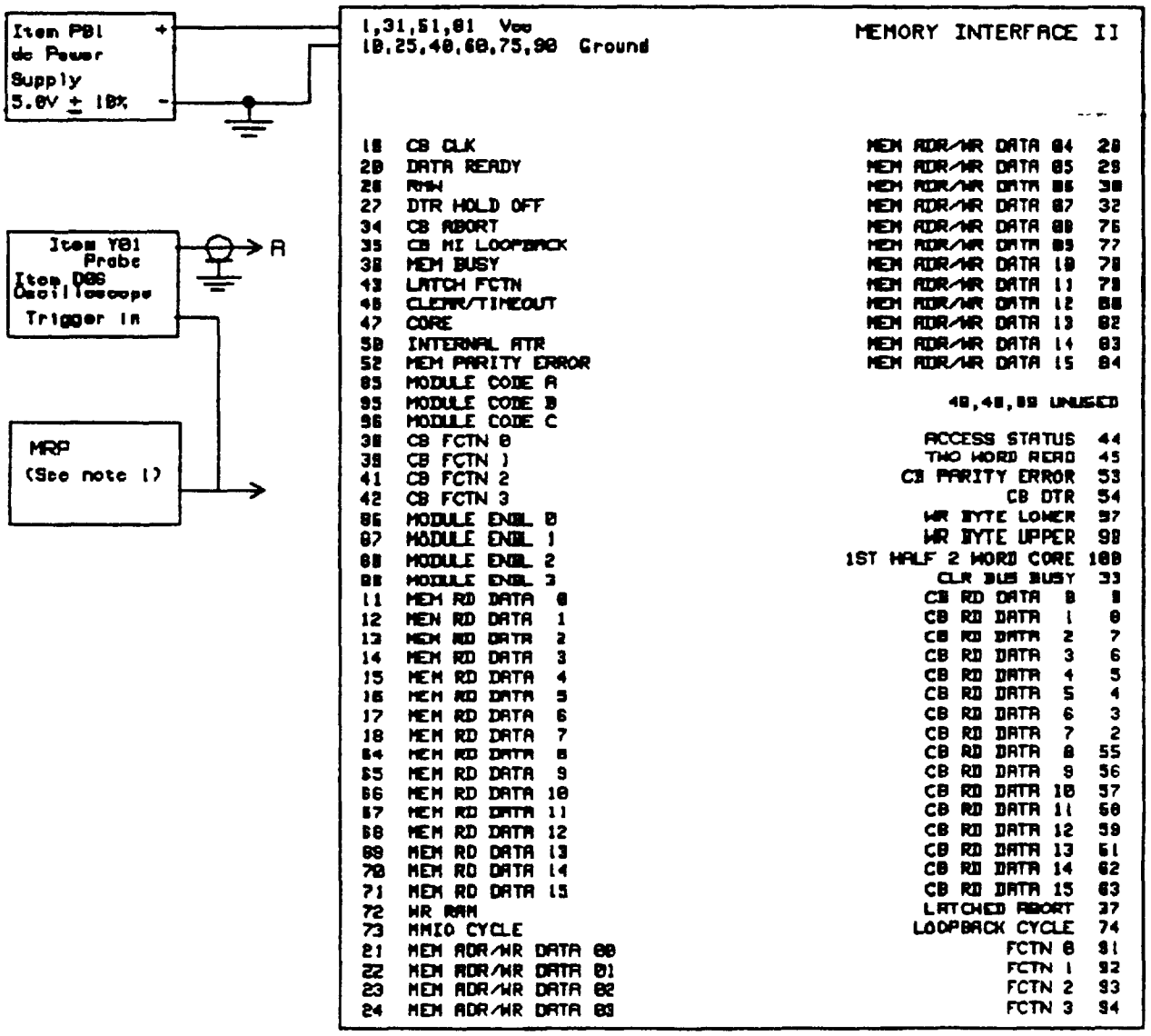


FIGURE 14. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

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MIL-M-28787/536	Modules, Standard Electronic, Memory Interface II Module, Key Code EWD

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
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